أكاديمية الفضاء الوطنية
NATIONAL SPACE ACADEMY

وكالة الإمارات للفضاء
UAE SPACE AGENCY



Intro to remote sensing fundamentals

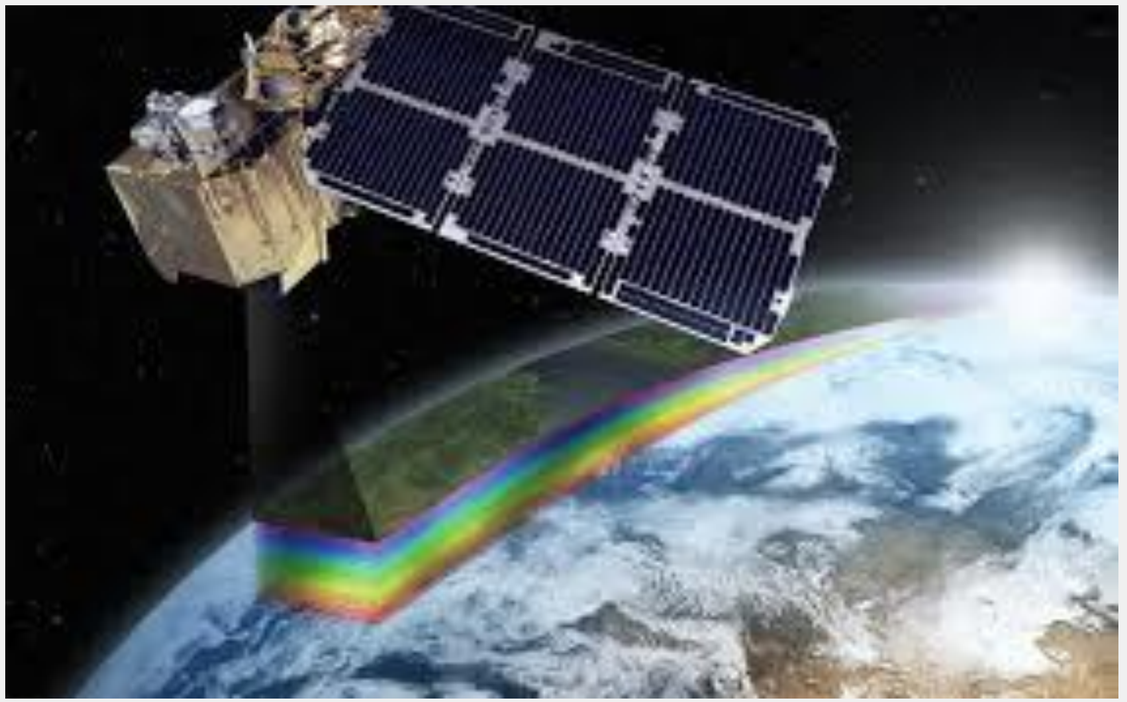
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Outline

Outline

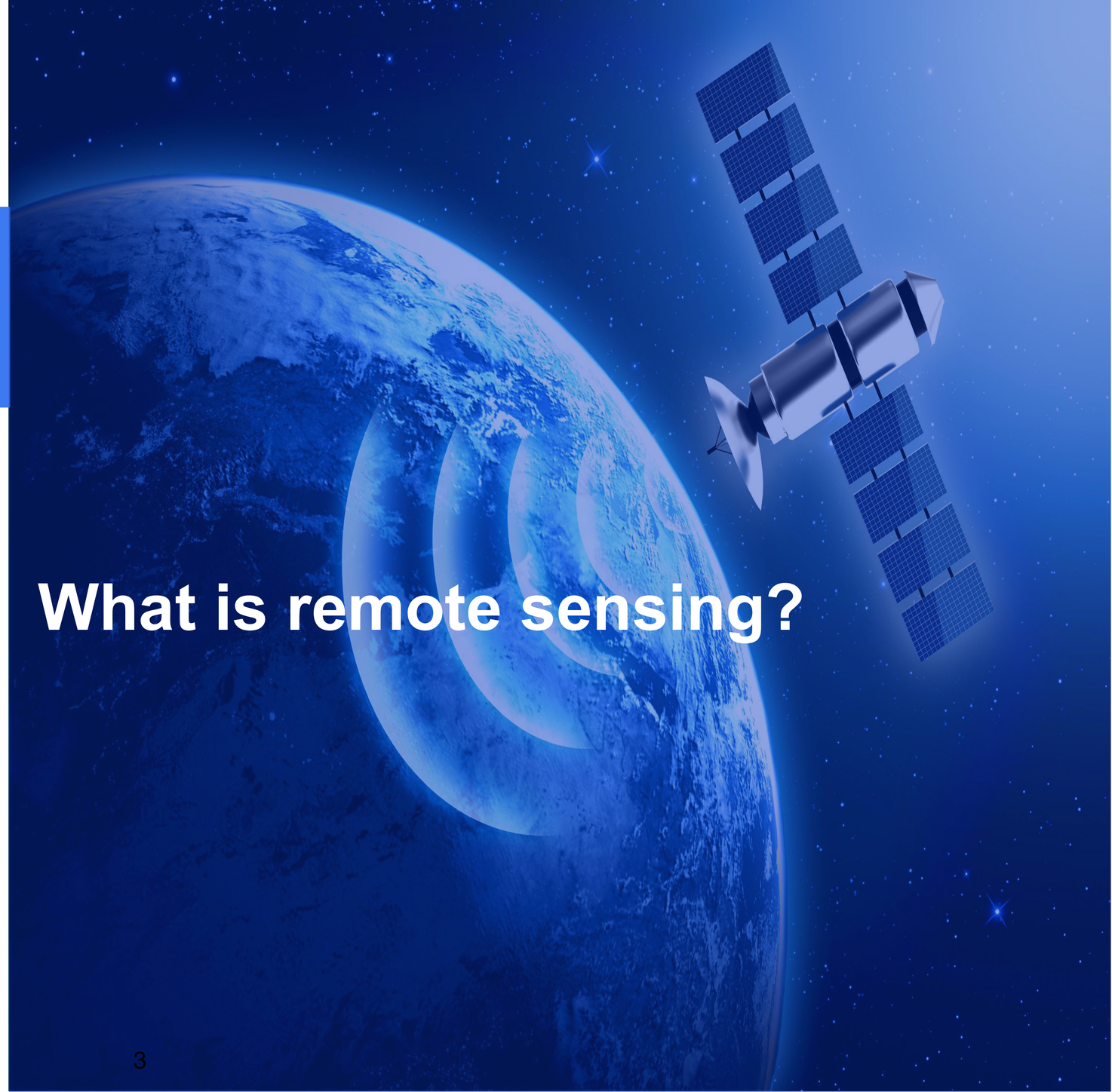
- What is remote sensing?
- Why remote sensing matters?
- Sensors and satellites
- Radiative transfer theory
- Data characteristics
- Advantages and disadvantages of RS
- Applications
- Projections and coordinate systems
- Summary



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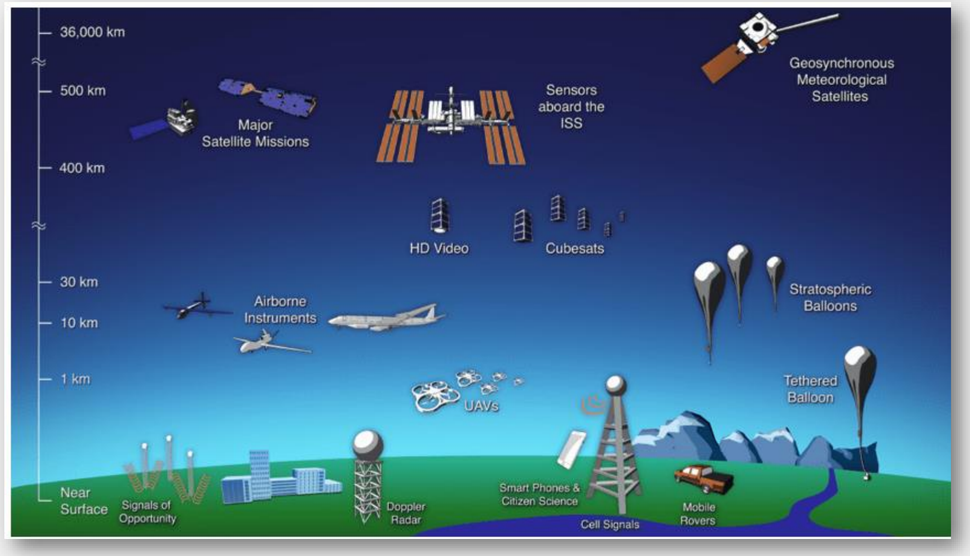
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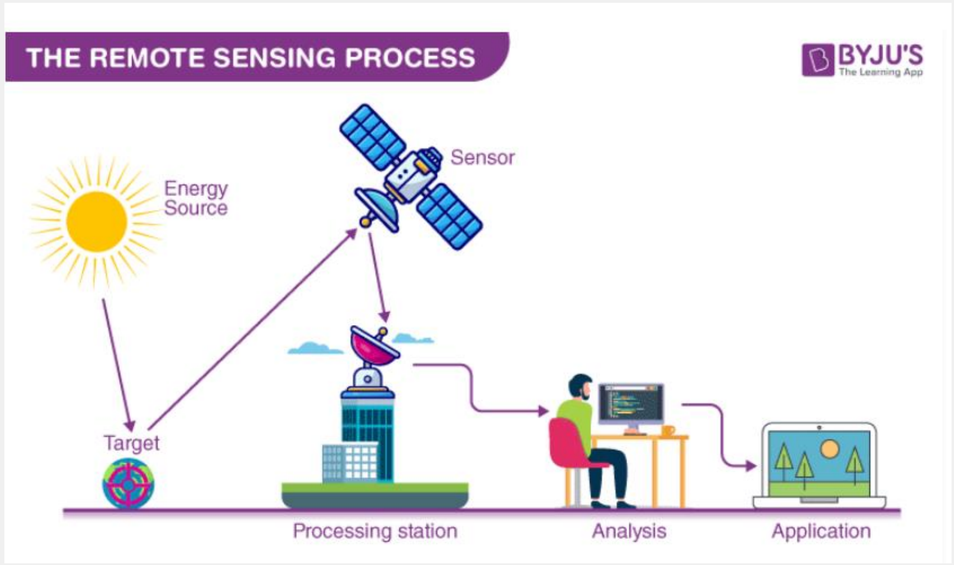
What is remote sensing?



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What is remote sensing?





Remote sensing: Obtaining information about an object from a distance

Photogrammetry is a very common form of remote sensing

Different ways to collect data and different sensors:

Space borne sensors
- Satellites

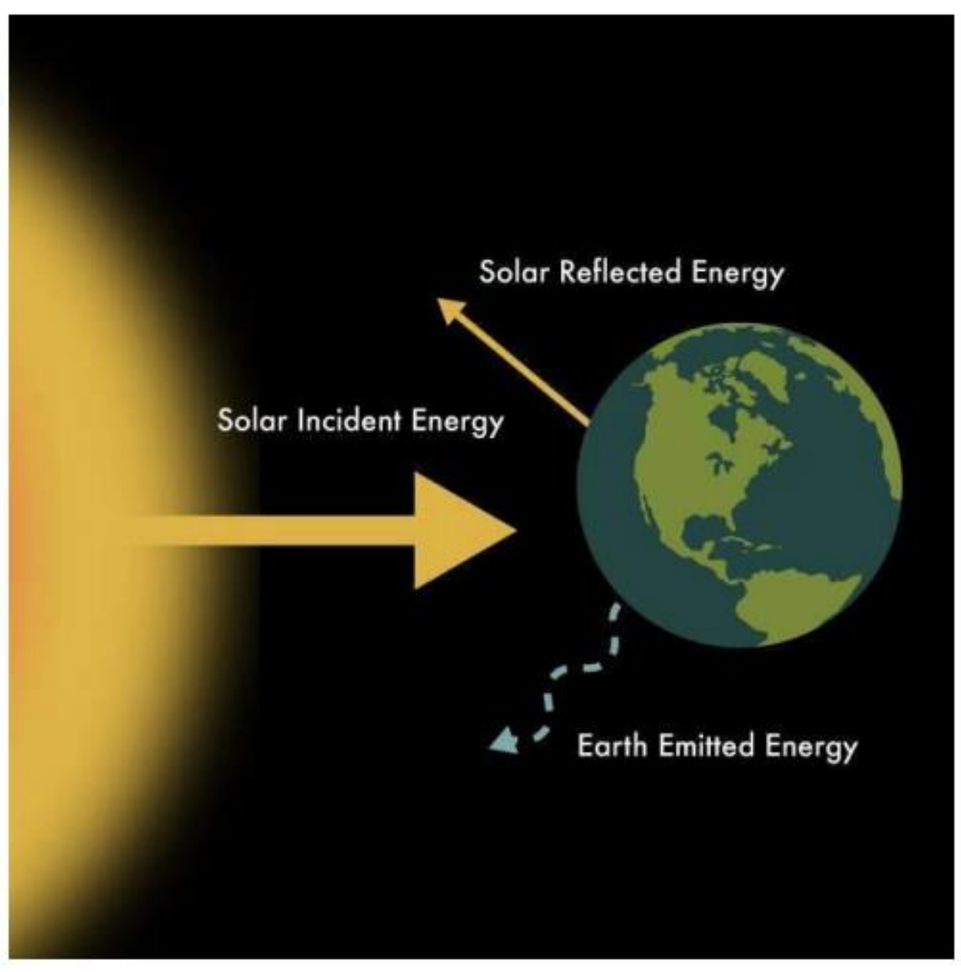
Airborne
- Airplane or drone

Ground based sensors
- e.g. Field spectrometers

- What information do we want to acquire?
- How detailed should this info be?
- How often do we need the data?

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What is remote sensing?



- Electromagnetic radiation = Energy received from the sun
- The earth's atmosphere affects the incident energy from the sun

- Satellites have sensors that monitor the electromagnetic radiation

- We install sensors on satellites to measure the radiation reflected or emitted from planets and atmospheres

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Remote sensing fundamentals

- Study of wavelength composition of electro magnetic radiation
- Electro magnetic radiation (EMR) is energy in electro magnetic waves
- Visible light, microwave, xray
- EMR can be transmitted, absorbed, or refracted

Refraction

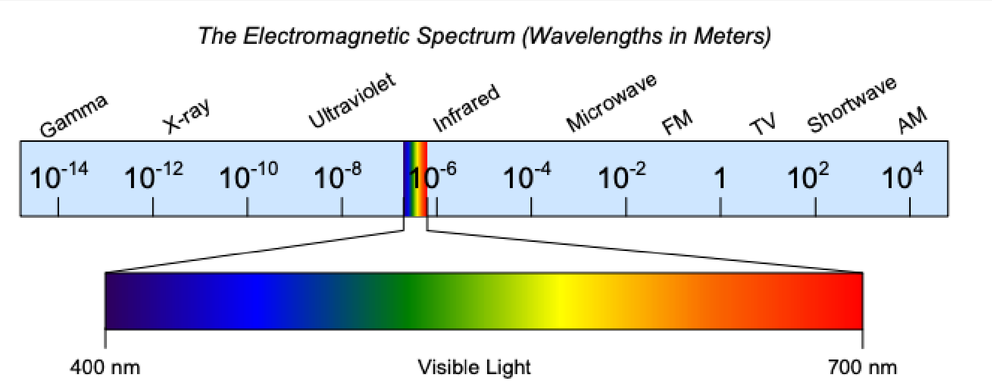
- When EMR passes through a substance, waves change direction/bend

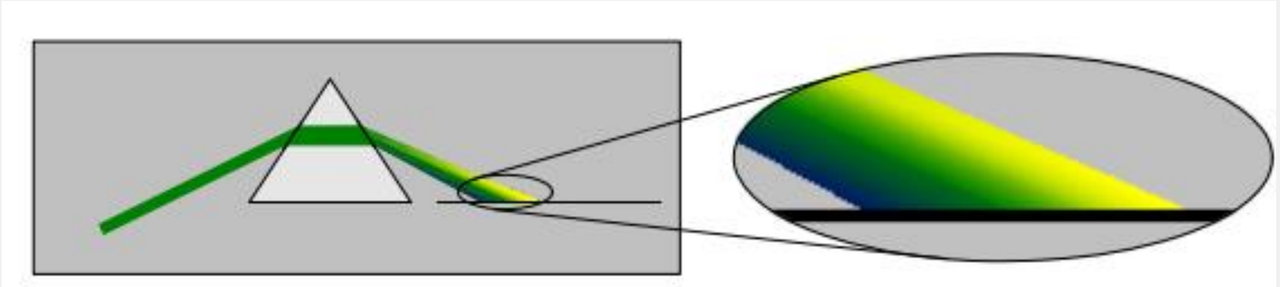
Visible light comprises of many bands

- White light is a mix of all colors

Natural white light is a mixture of wavelenghts

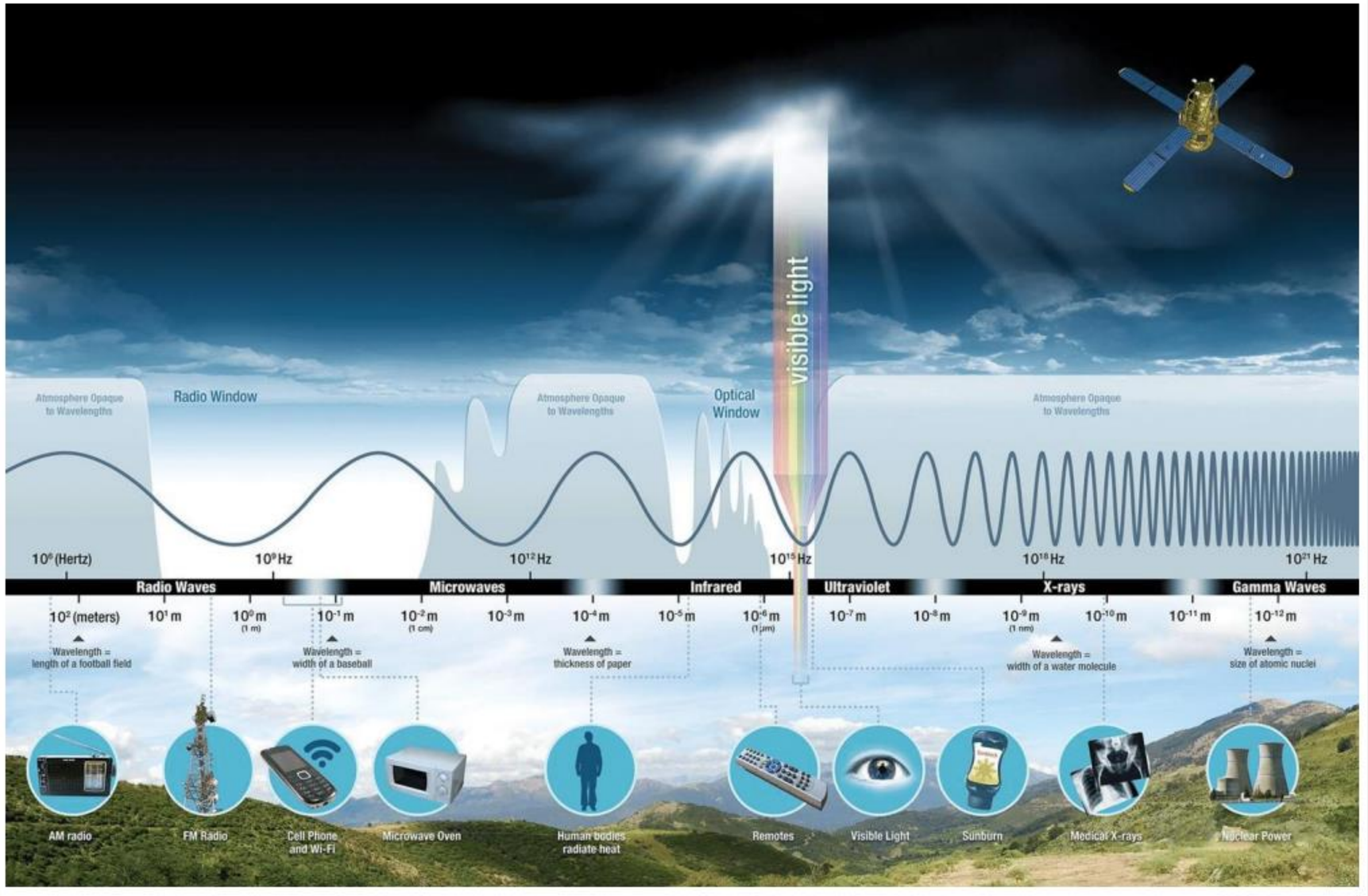
- E.g. green light from a tree is a mixture of blue, green and yellow light





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Electrtro magnetic spectrum



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Why remote sensing matters?



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Why remote sensing matters?

Global perspective at scale 🌎

- Satellites provide a bird’s-eye view of Earth
- Allows for global monitoring of our natural ecosystems: oceans, forests, ice caps, urban growth. Ground-based sensors cannot achieve alone

Environmental monitoring and climate change 🌴

- Track changes in forest cover, rising levels, and more
- Scientists use satellite data to understand, model and predict climate change

Disaster response & risk reduction 🚨

- Remote sensing provides real-time data for rapid response
- Helps to save lives and prepare for future events



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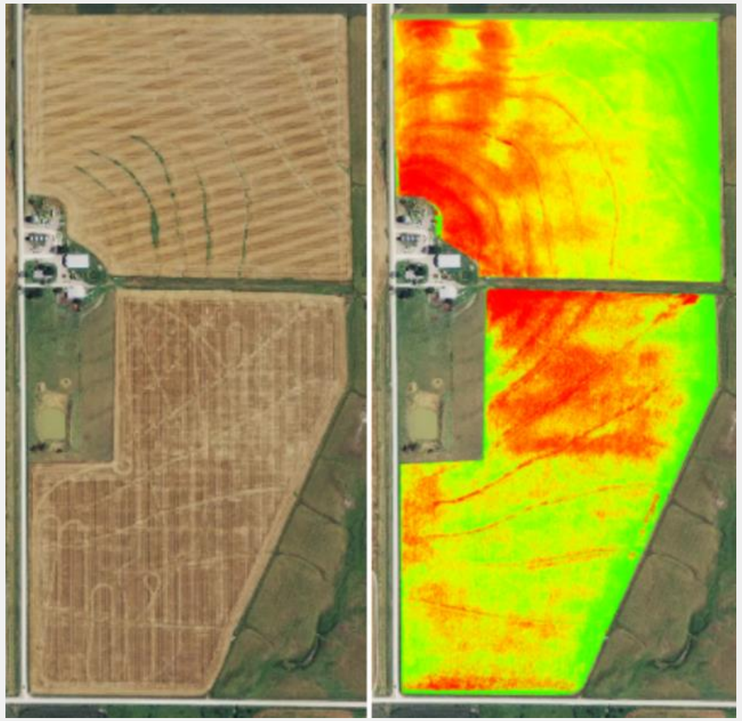
Why remote sensing matters?

Food security and agriculture 🌽

- Satellites monitor crop health, soil moisture, drought conditions and land use changes
- Improves farming efficiency and food supply chain planning

Water resource management 💧

- Track soil moisture changes due to rising surface temperature
- Monitor water quality of marine ecosystems

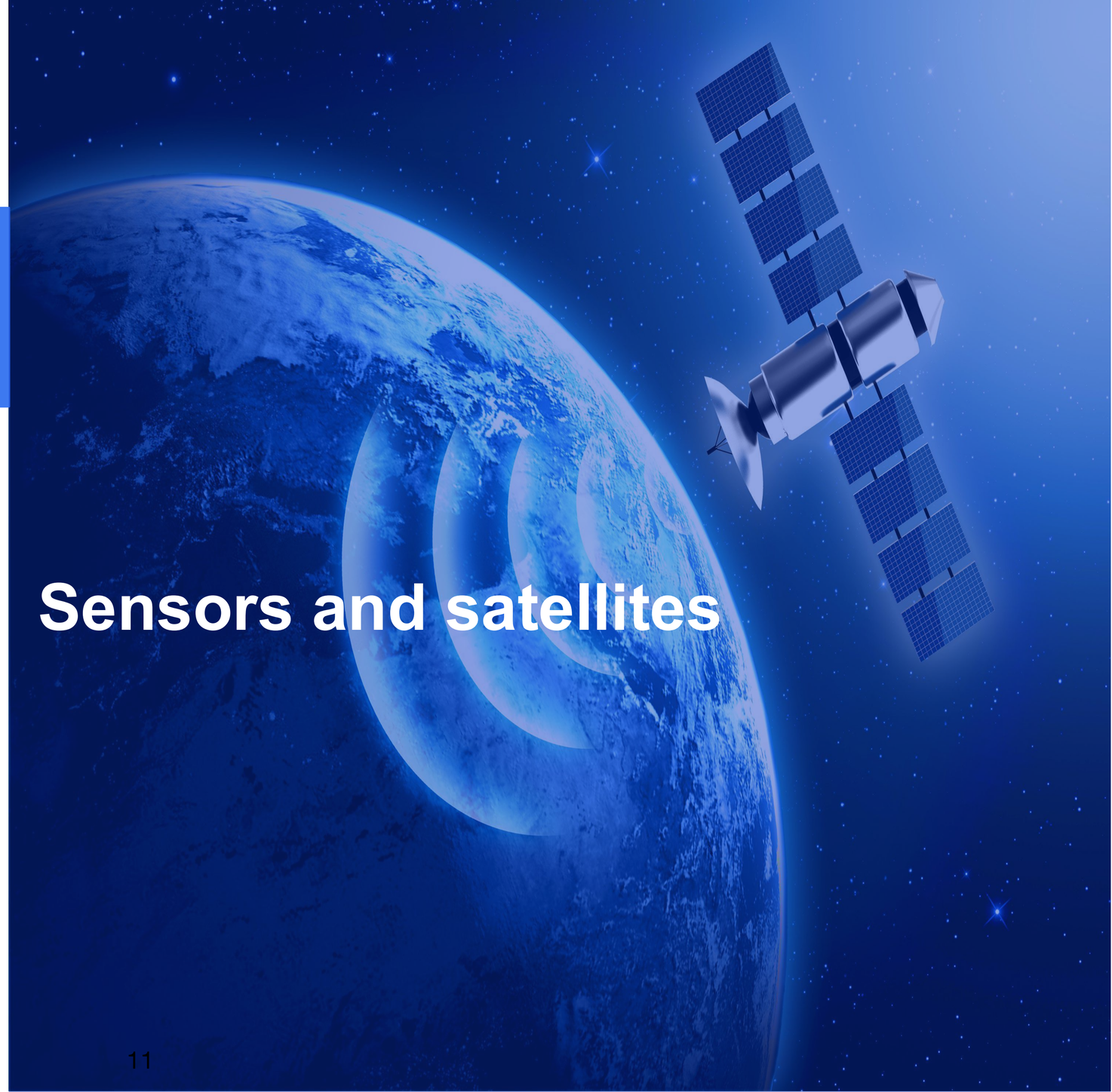


Space-based remote sensing empowers humanity with a tool to **observe**, **understand, and manage earth.**

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Sensors and satellites



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Satellite characteristics

Satellite Characteristics

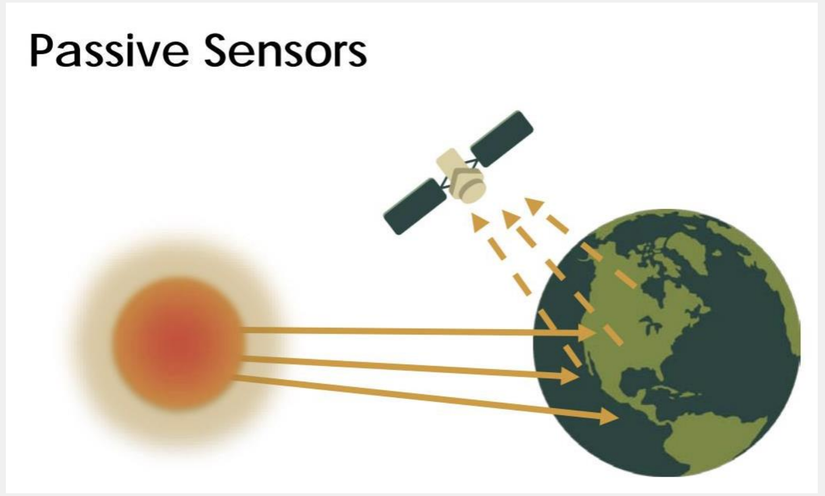
- Orbits: Polar/Non-Polar Orbit vs. Geostationary
- Energy Source: Passive vs. Active
- Solar and Terrestrial Spectra: Visible, UV, IR, Microwave...
- Measurement Technique: Scanning; Non-Scanning; Imager; Sounders
- Resolution Type and Quality: Spatial, Temporal, Spectral, Radiometric
- Application: Weather, Ocean Color, Land Mapping, Air Quality, Radiation Budget, etc.

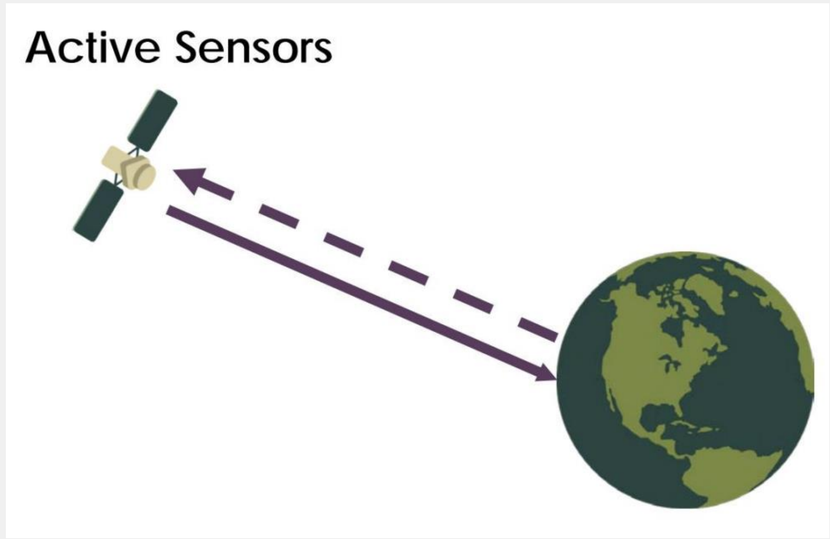
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Satellite characteristics: sensor types

Commonly we have 2 main types

- Active: uses an active signal, laser (LiDAR) or Radar
- Passive: optical RS uses the sun as an energy source



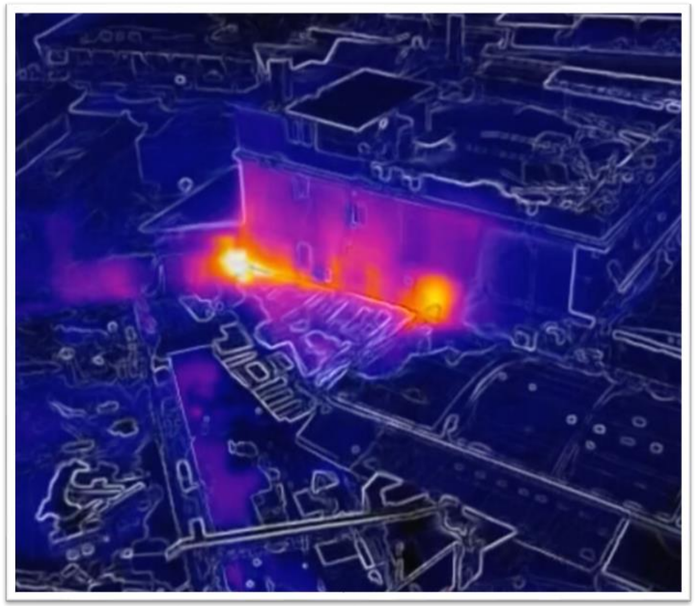


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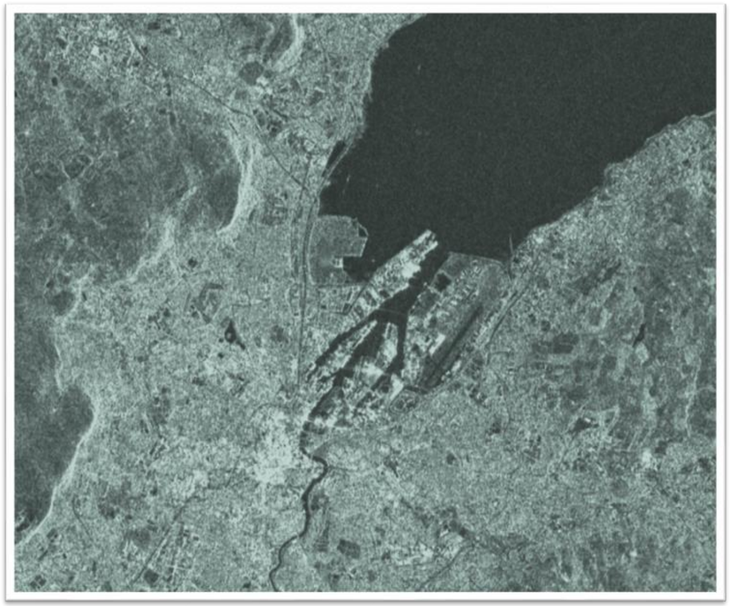
Examples of data used in remote sensing



Optical Data



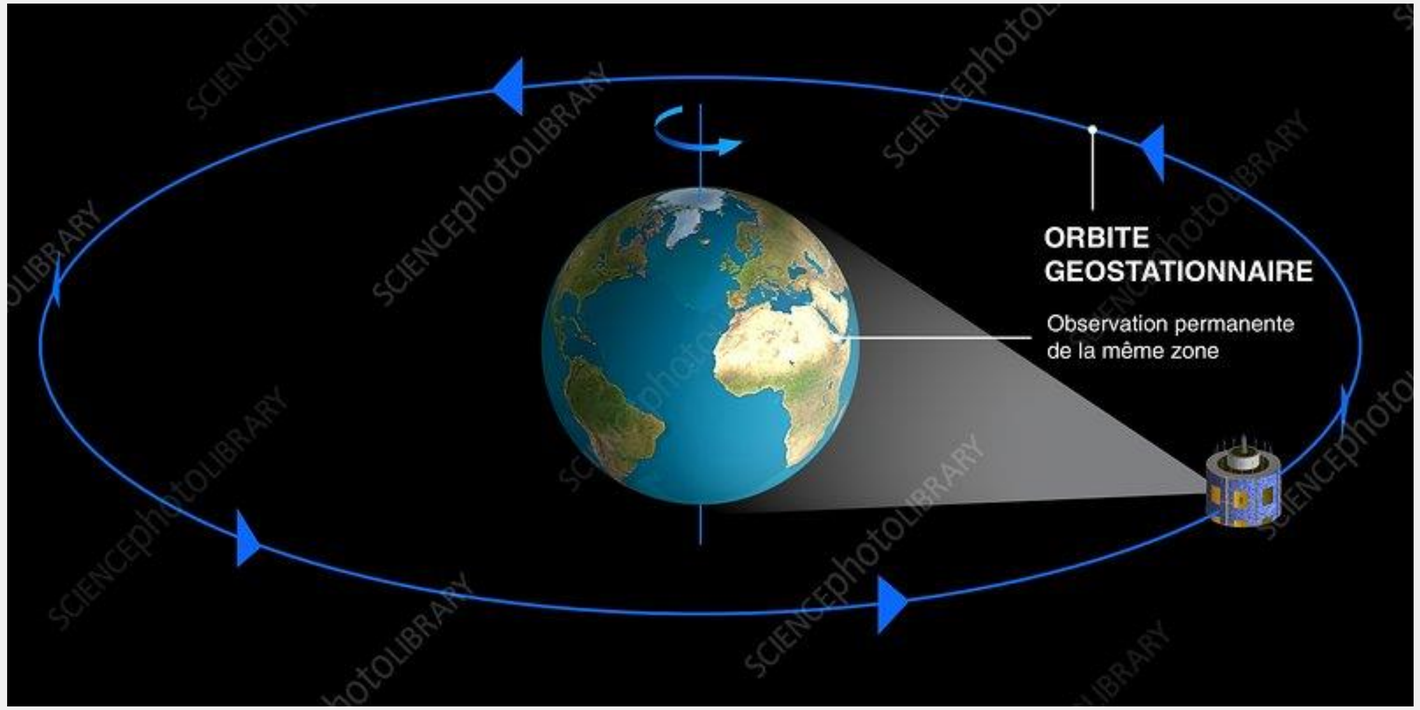
Thermal Infrared Data



Microwave Data (SAR)

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Satellite characteristics: orbits

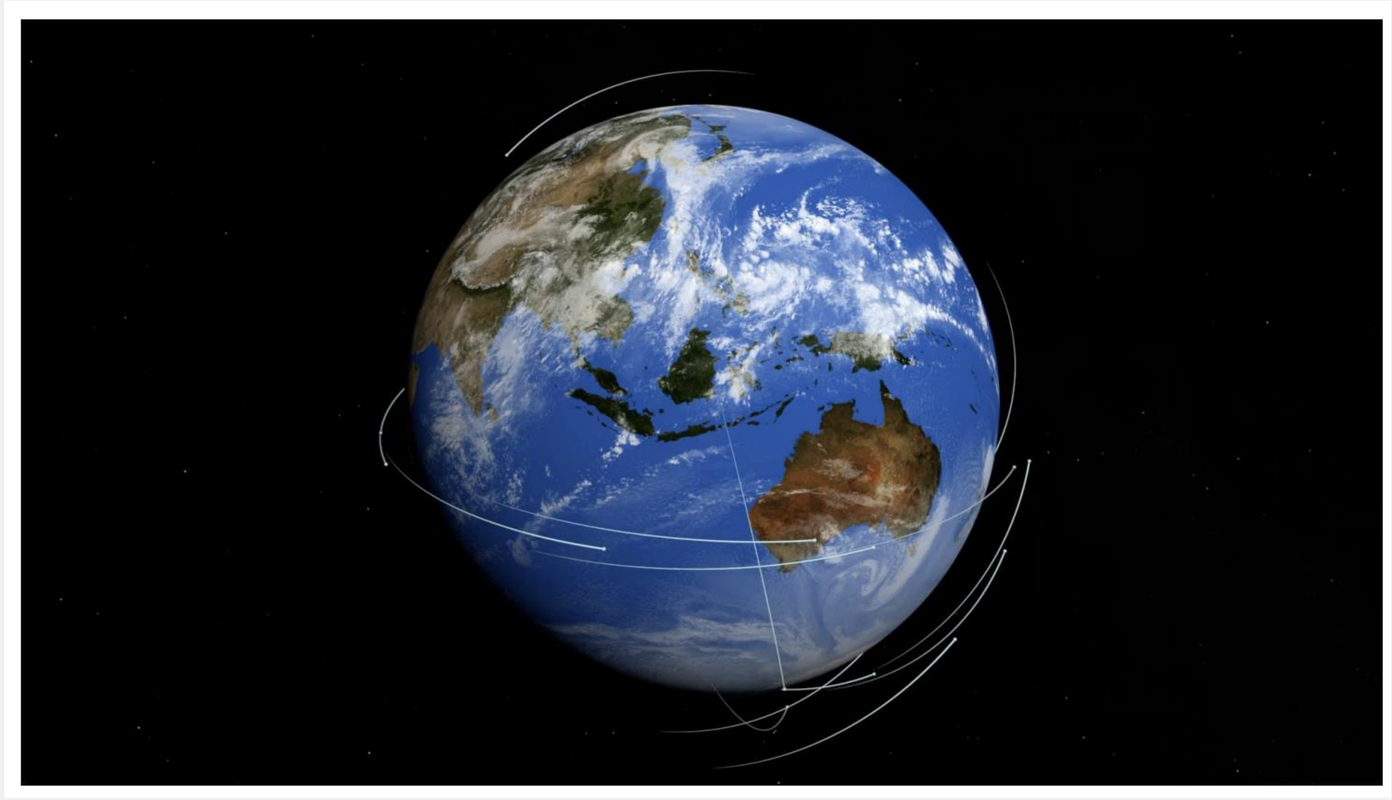


Geostationary orbits

- Satellites fly around the equator at ~ 36 000 km with the same speed as the earth's rotation. Thus becoming 'stationary''
- Always measure the earth's sunny side
- Cover a huge area with low resolution
- Great for weather or communication applications

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Satellite characteristics: orbits

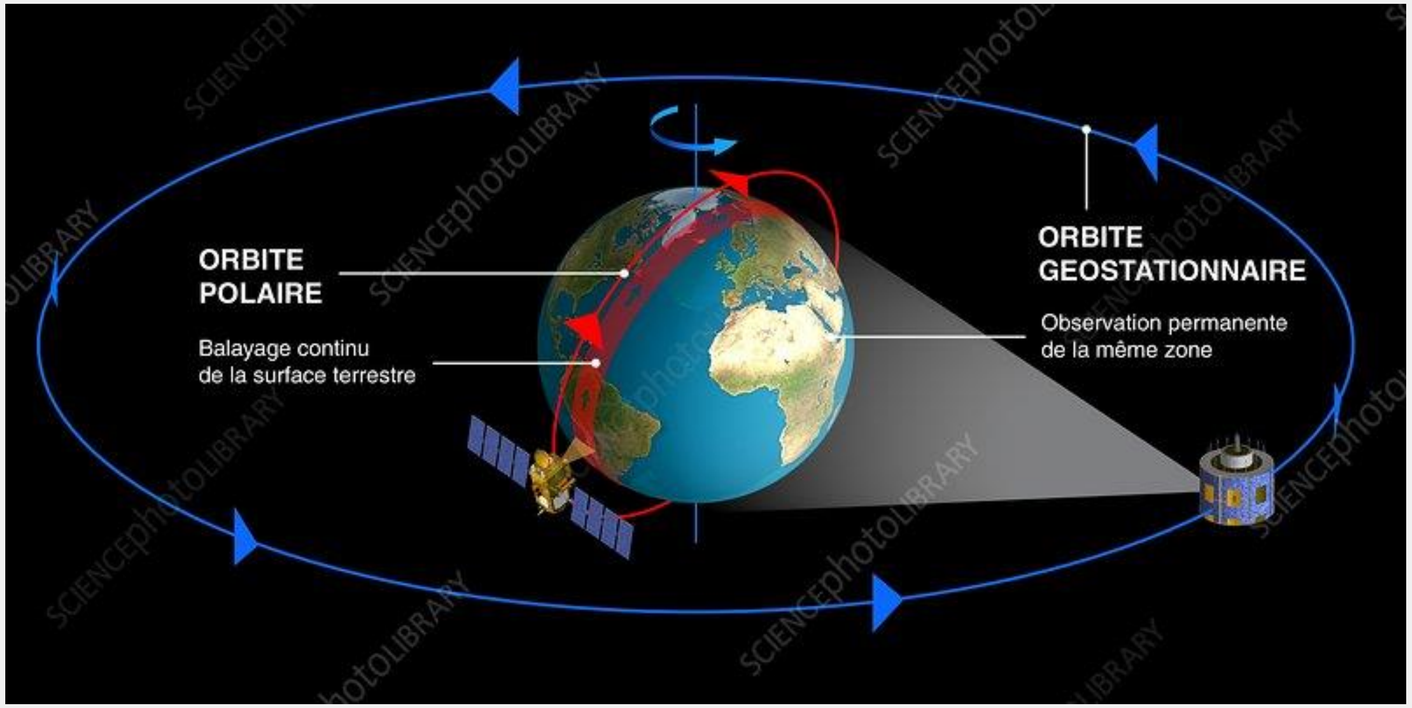


Low orbit (LEO)

- Satellites flying in lower orbits around the earth
- Can fly from pole to pole
  - NASA’s landsat satellite
- Non Polar like the ISS or GPM
- Less frequent measurements
- Medium pixel resolution 10m – 100m
- Multiple satellites cover almost entire earth

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Satellite characteristics: orbits



Polar orbit

- Build for global coverage
- Measurements can range from one per day to one a month
- High revisit time for same area
- Sun synchronous = always flies over same area at solar noon = 12 am.
  - Why would that be?

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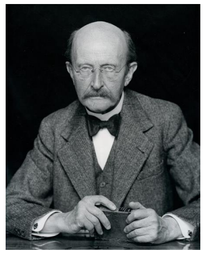
Radiative transfer theory

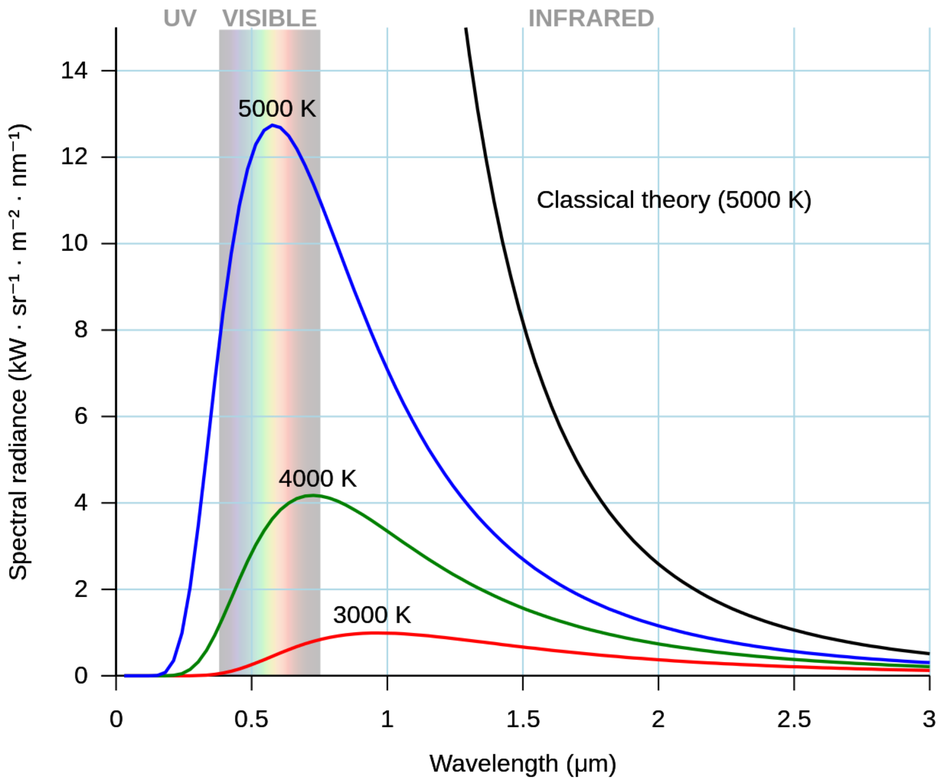


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Thermal physics law







Planck's law:

- Imagine a metal pipe you heat in the fire

Planck's law tells us how much light (EMR) is emitted at different wavelengths depending on the temperature.

Stefan-Boltzmann law:

Bigger fire emits more radiation = more heat

Boltzmann law: If you double the temperature of an object the emitted energy increases 16 times (2^4).

Wien's displacement law

Flame color changes with temperature

Wien's law tells us the color's peak emission shifts with wavelength as temperature changes.

e.g. blue flame = emits blue light = hot flameRed flame = emits more red light = less hot flame



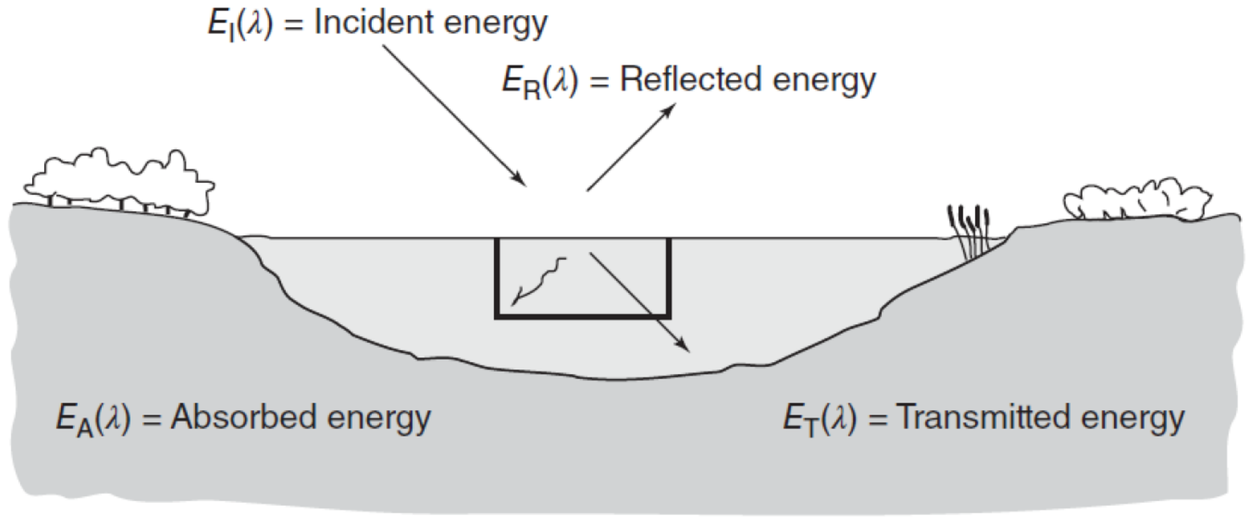


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Radiative transfer theory

Various **fractions** of the energy incident on the element **are reflected, absorbed, and/or transmitted**

$$E_I(\lambda) = E_R(\lambda) + E_A(\lambda) + E_T(\lambda)$$



$$\rho(\lambda) = \frac{E_R(\lambda)}{E_I(\lambda)}$$ Reflection coefficient

$$\tau(\lambda) = \frac{E_T(\lambda)}{E_I(\lambda)}$$ Transmission coefficient

$$\alpha(\lambda) = \frac{E_A(\lambda)}{E_I(\lambda)}$$ Absorption coefficient

[10] K. Tempfli et al.

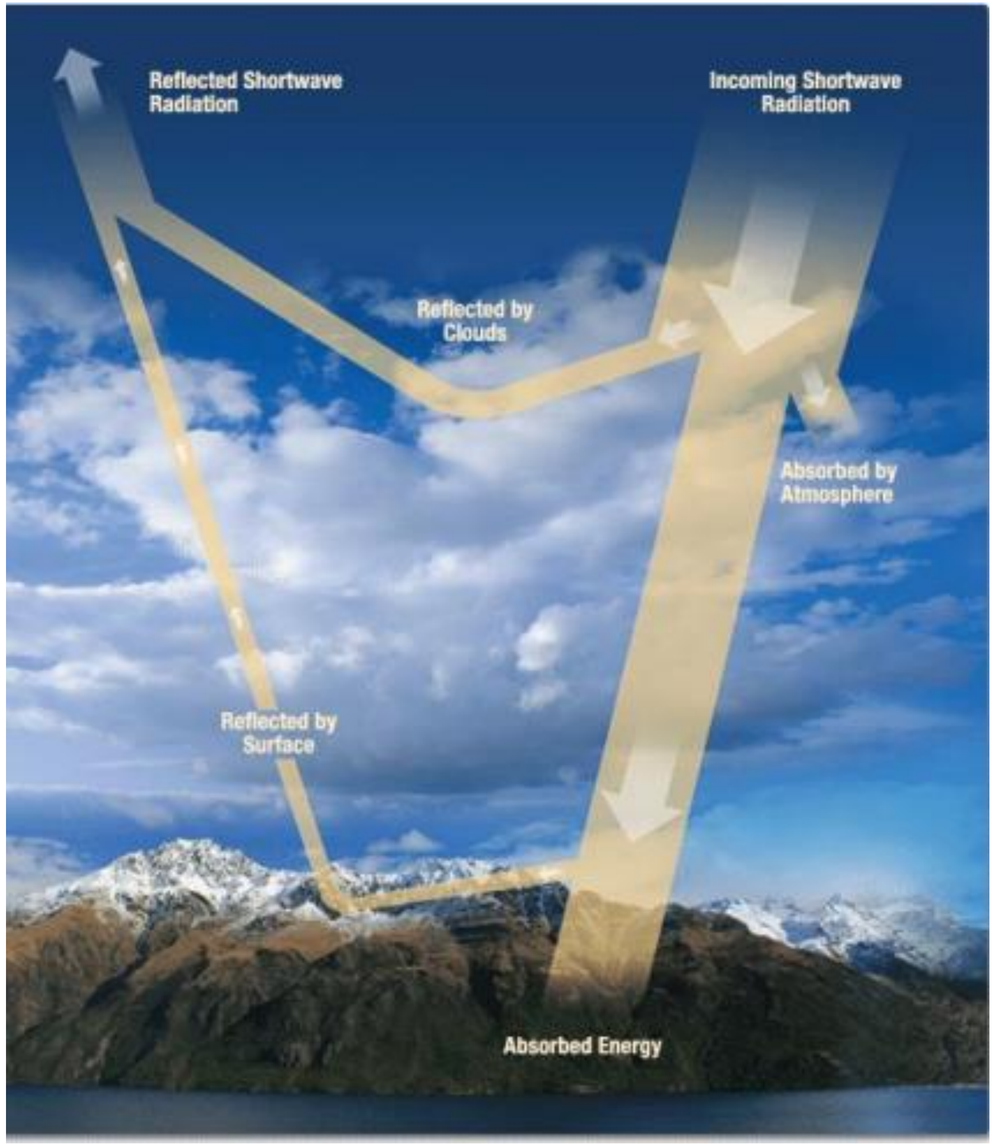
From the **conservation of energy** theorem, it can be derived

$$E_I(\lambda) = E_R(\lambda) + E_T(\lambda) + E_A(\lambda)$$

$$\rho(\lambda) + \tau(\lambda) + \alpha(\lambda) = 1$$

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Atmosphere



- Radiation passes through the atmosphere twice, from the sun to the sensor and back.
- Aeorosol particles and gases in the atmosphere absorb most of the incident energy before it reaches earth.
- Hence we apply atmospheric correction to all satellite image data to remove the atmospheric scattering and absorption effects . Then we can obtain the surface reflectance properties.

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Data characteristics



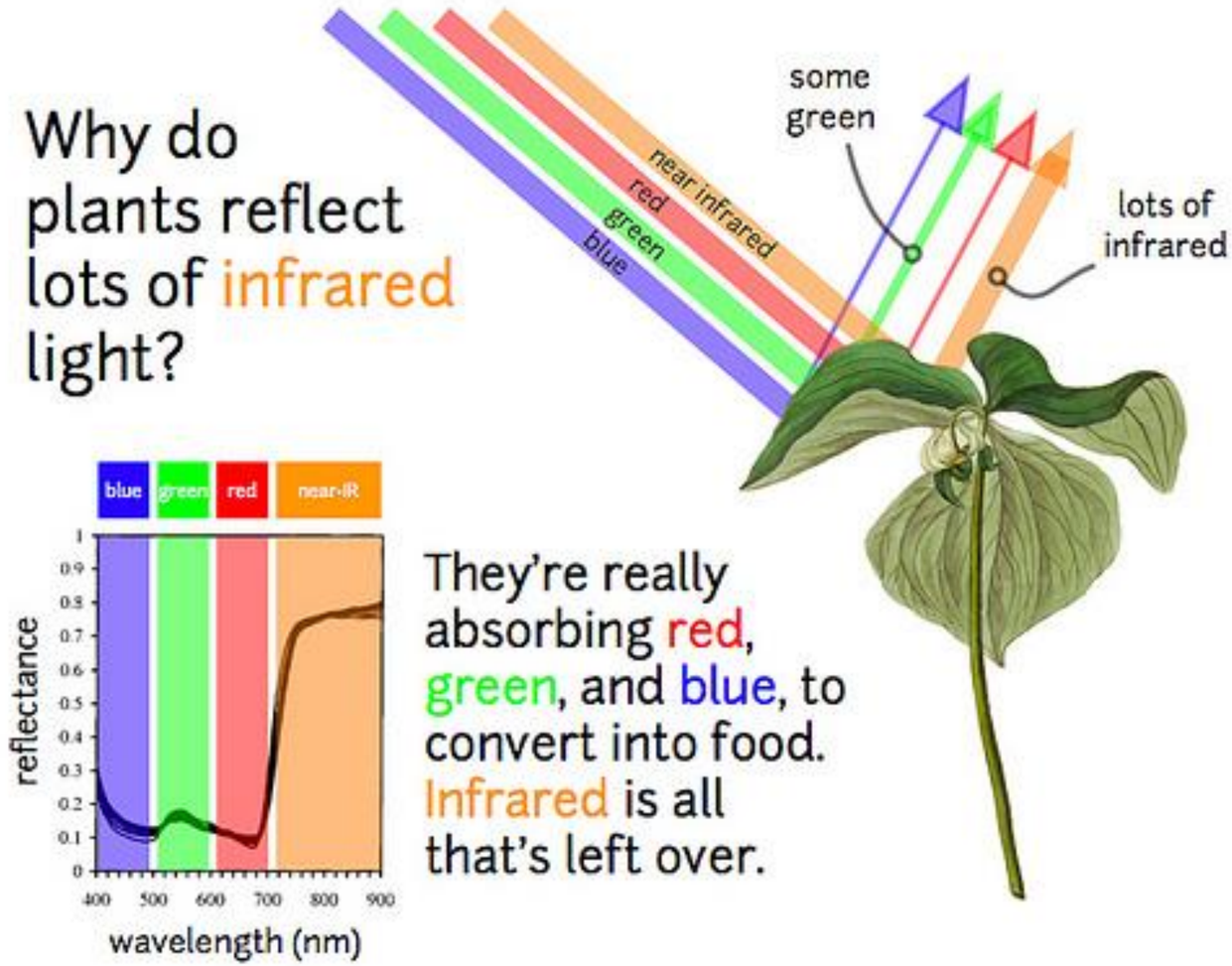
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Spectral signature. What's that?

- Spectral reflectance
  - Proportion of incoming vs reflected energy

$$E_R(\lambda) = E_I(\lambda) - [E_A(\lambda) + E_T(\lambda)]$$

  - Function of wavelength

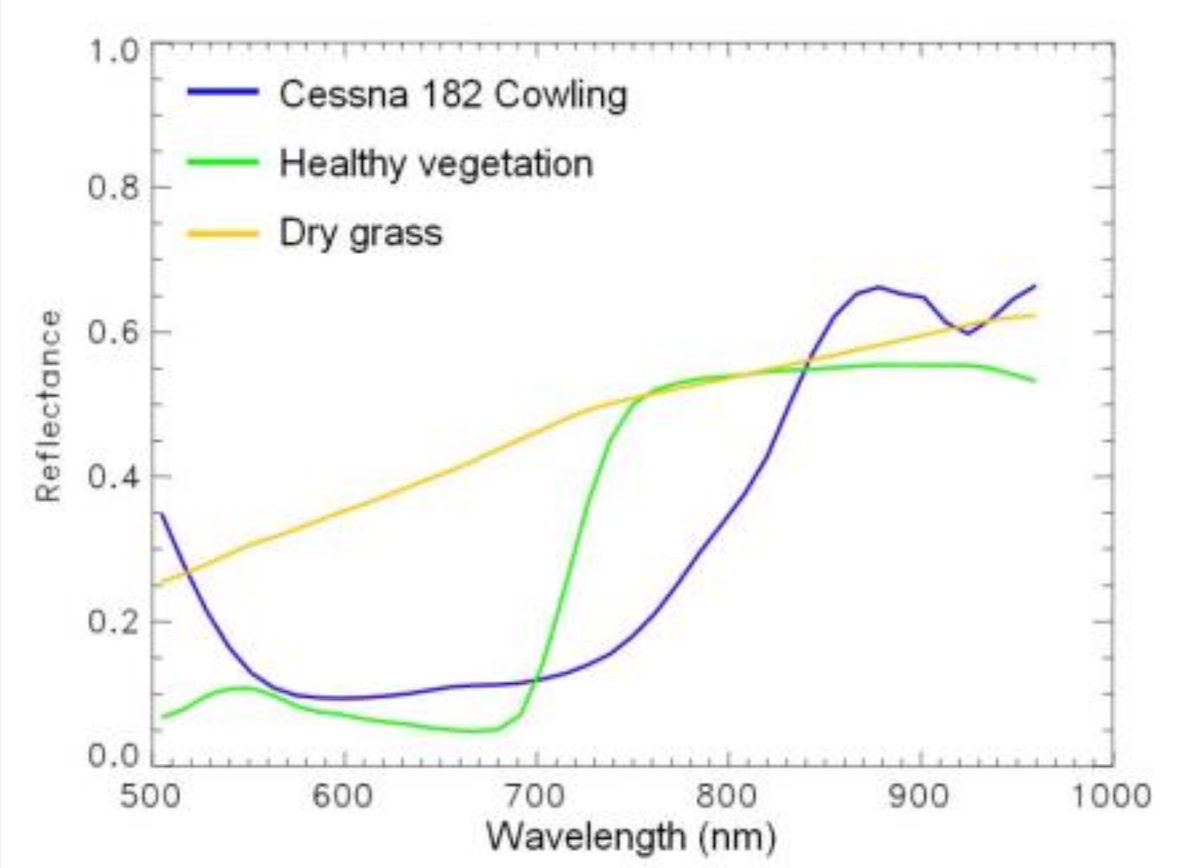


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- Using light refraction we can collect unique intensity signatures
- The refracted light can be used to discriminate similar looking objects
- Reflectance is a physical property
- % of reflected incident EMR per wavelength
- Unique fingerprint

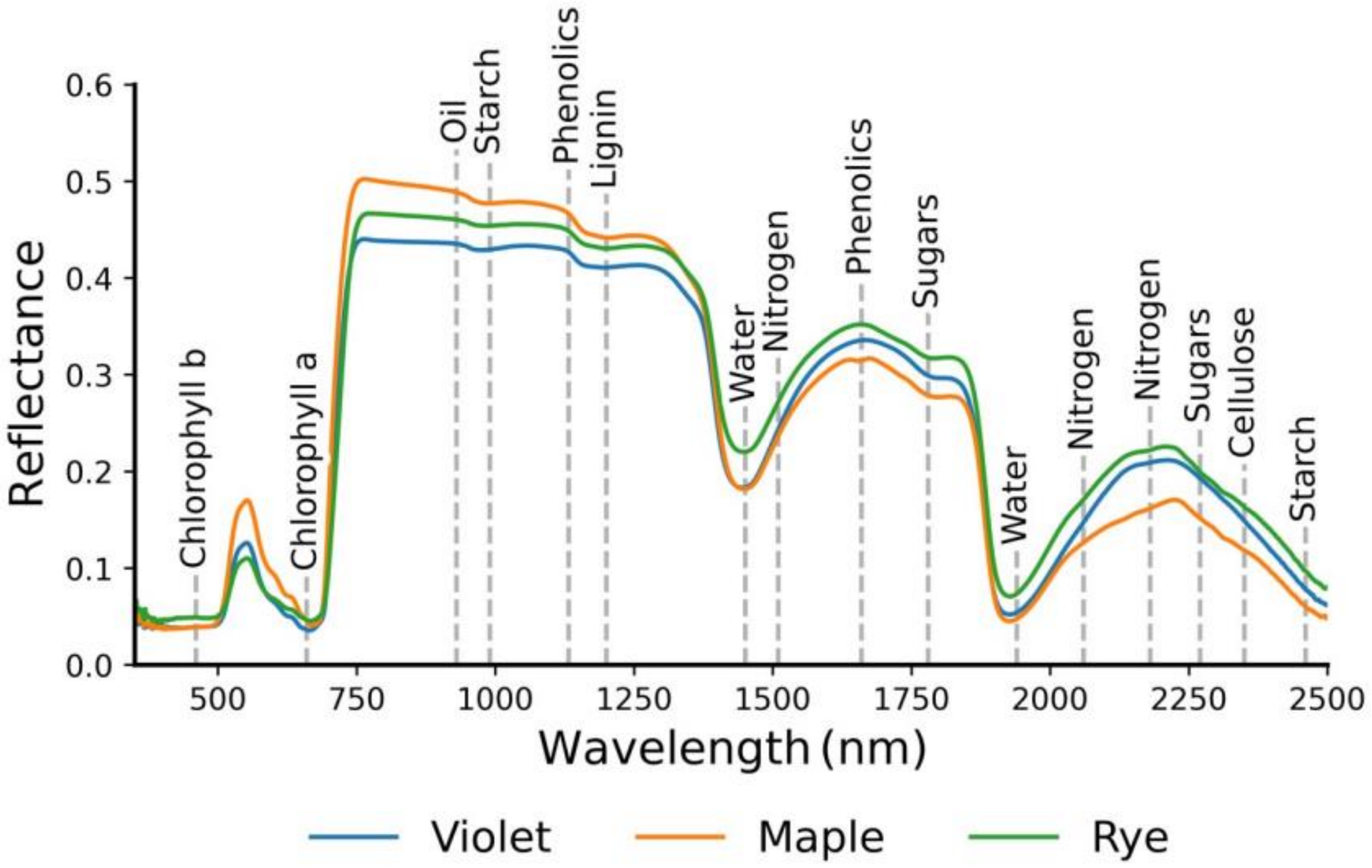
- Spectral signature depends on:

1. The source that illuminates the object
2. The reflectance of the object



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Reflectance Spectroscopy

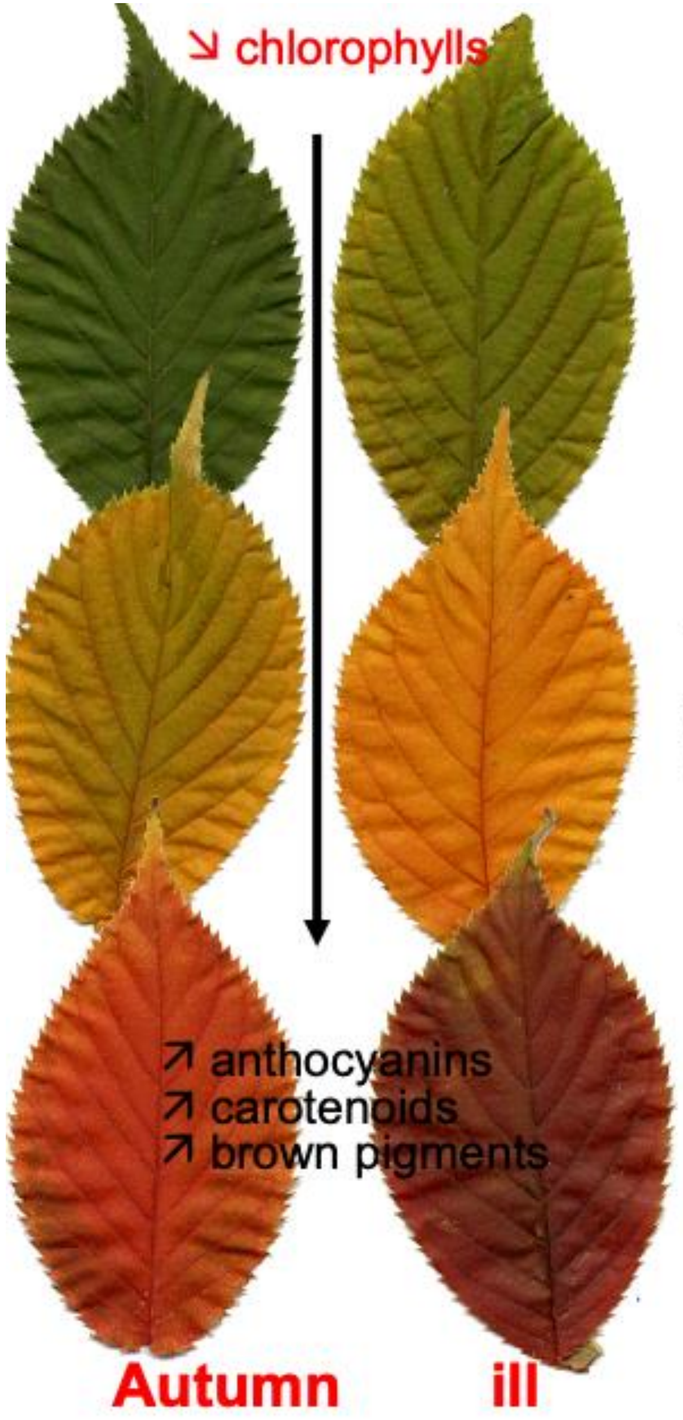




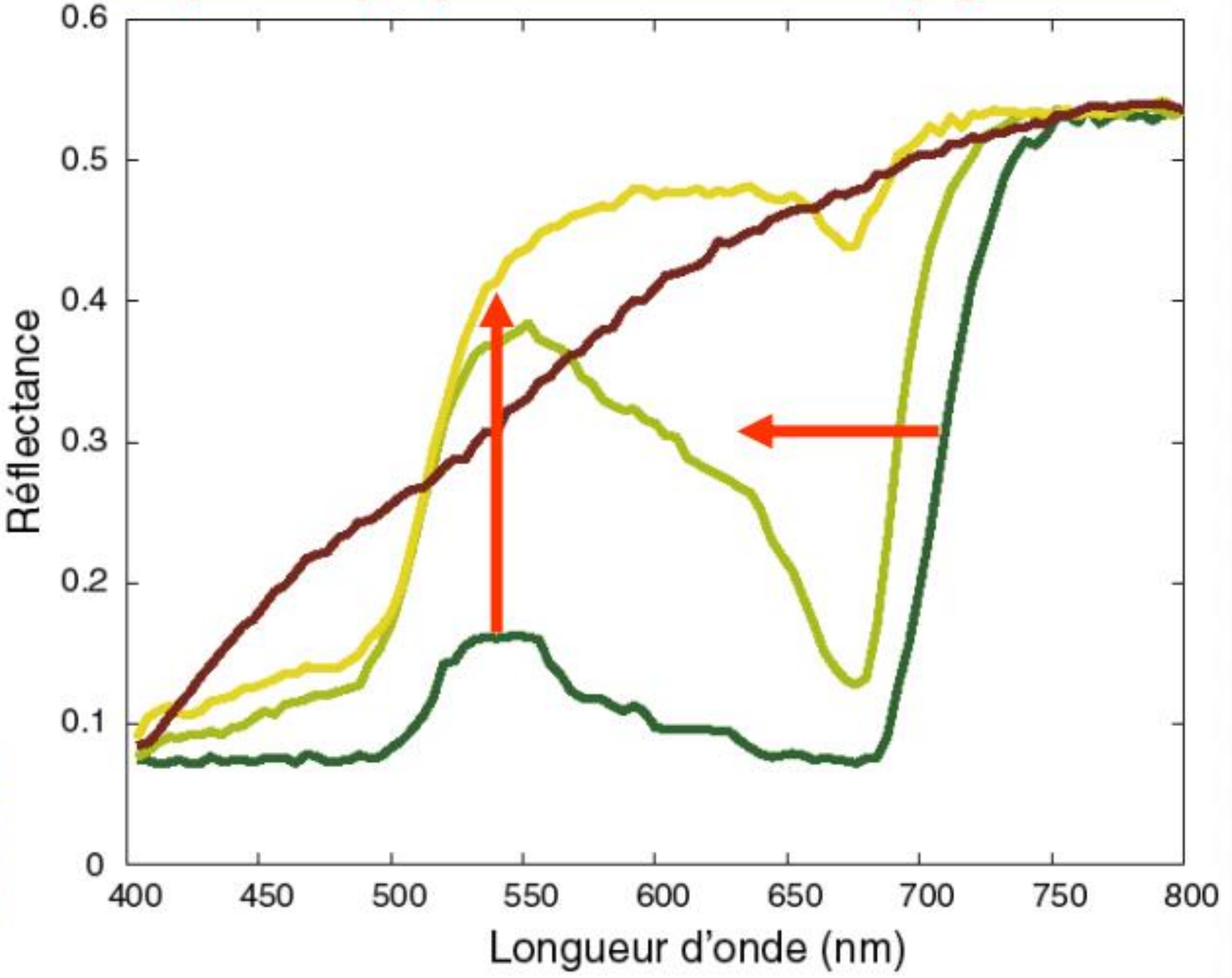




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Spectral properties: effect of leaf pigments



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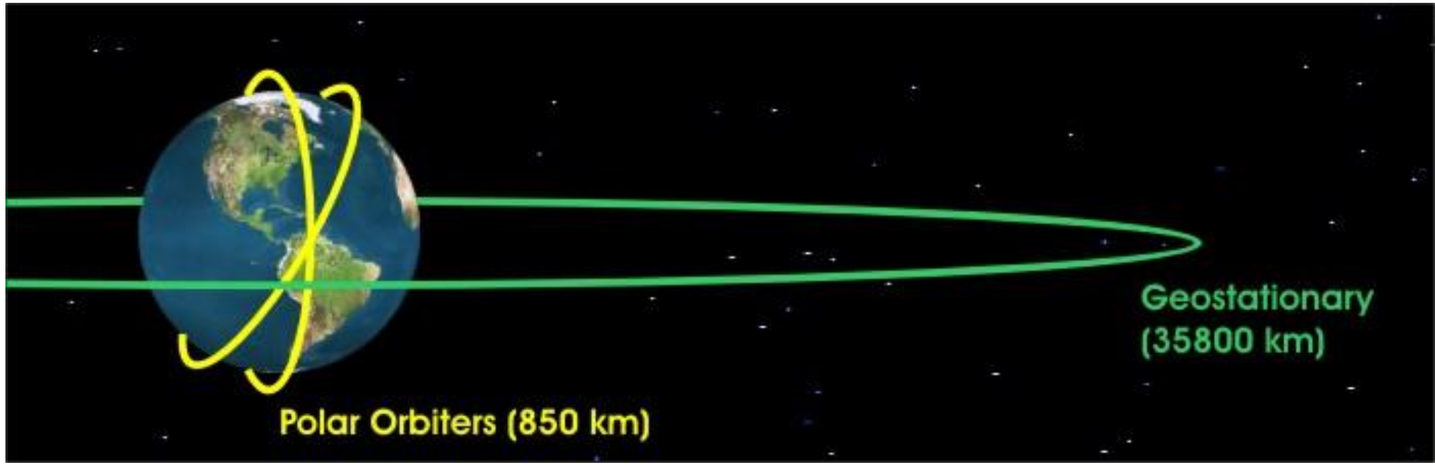
Data characteristics

Data quality:

- Spatial resolution
  Size of a pixel: cm – km

  Implies what is the smallest object on ground.

- Temporal resolution
  Acquisition time interval: daily, bi-weekly, monthly, annual



Spectral resolution

Wavelength range: visible range (400-700nm), NIR (700-1200nm), Mid infrared, thermal etc.
Number of spectral bands
Width of the spectral bands

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Spatial resolution: deep dive

- Spatial resolution is influenced by the satellite's orbit and the design of its sensors, which vary across systems.
- It represents the ground area captured by a single pixel in the image; in some cases, features smaller than a pixel (sub-pixel objects) can still be detected.
- Typically, spatial resolution is expressed as the length of one side of a square pixel.
- Higher spatial resolution means each pixel covers a smaller area on the ground.
- The image in the bottom right illustrates how spatial resolution affects detail, with examples at 1 m, 10 m, and 30 m resolution from left to right.

Sensor	Spatial Resolution
DigitalGlobe (and others)	<1 m - 4 m
Landsat	30 m
MODIS	250 m - 1 km
GPM IMERG	~10 km

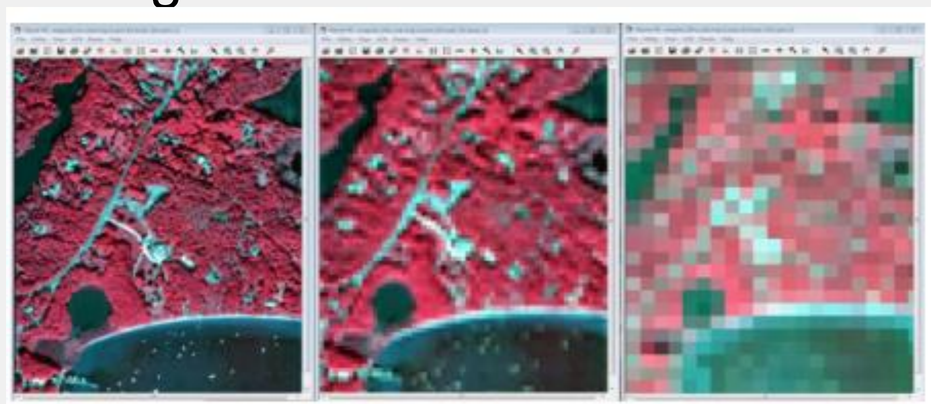


Image Credit: csc.noaa.gov

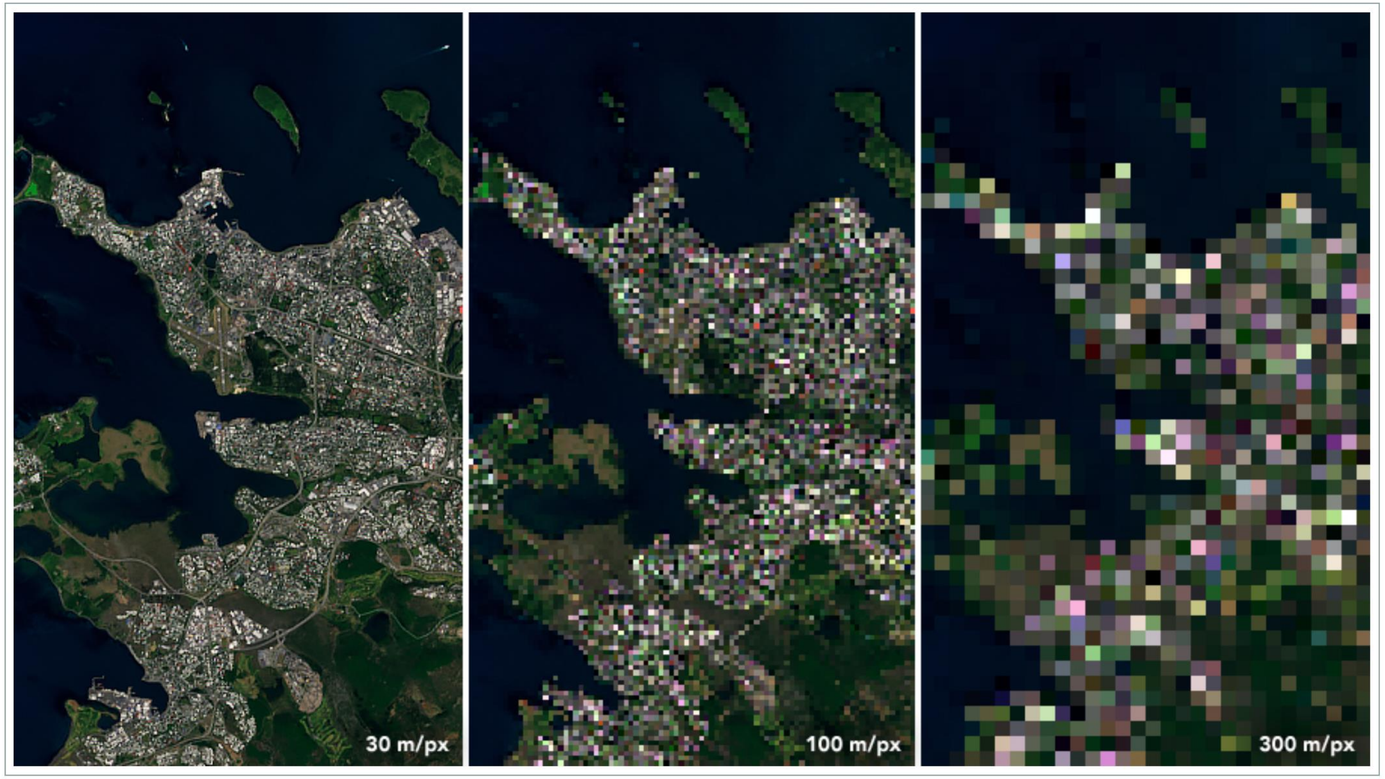
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Data characteristics

The objective defines the data quality that is needed

- Forest change detection: 30x30m spatial resolution, 4 images annually,
- Crop health monitoring: 2x2cm spatial resolution, bi-weekly
- Continental Land cover analysis: 1x1km spatial resolution, monthly.

Higher resolution is not always better!



Landsat 8 data from July 7, 2019 over Reykjavík, Iceland. Courtesy of NASA Earth Observatory.

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Temporal resolution

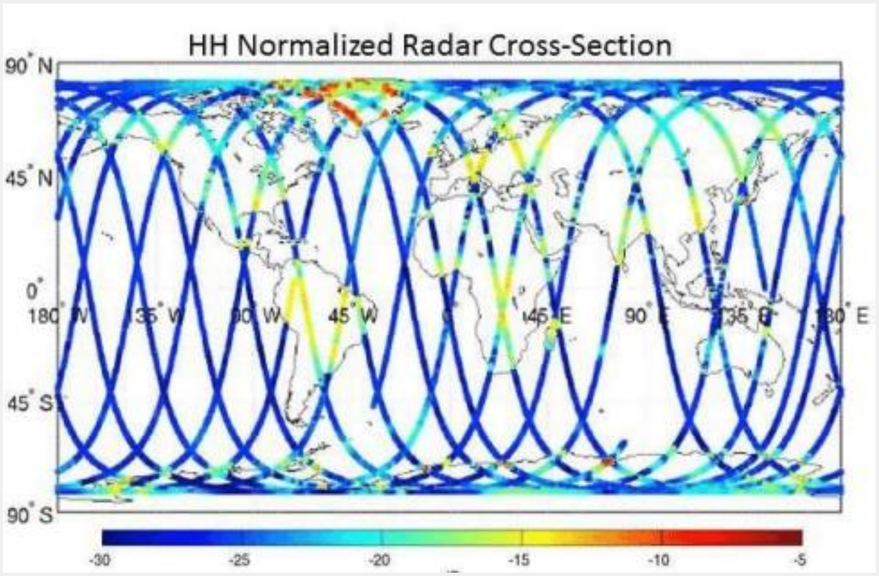
- The time it takes for a satellite to complete one orbit cycle. Also known as ‘revisit time’

- Satellite or sensor dependent, swath overlap or latitude

- Same satellites have greater temporal resolution because:

1. They can maneuver their sensor
2. Increasing overlap at higher altitudes

Sensor	Revisit time
Landsat	16-days
Modis	2-days
Sentinel-2	5 days
Commercial sensors	1-2 adys



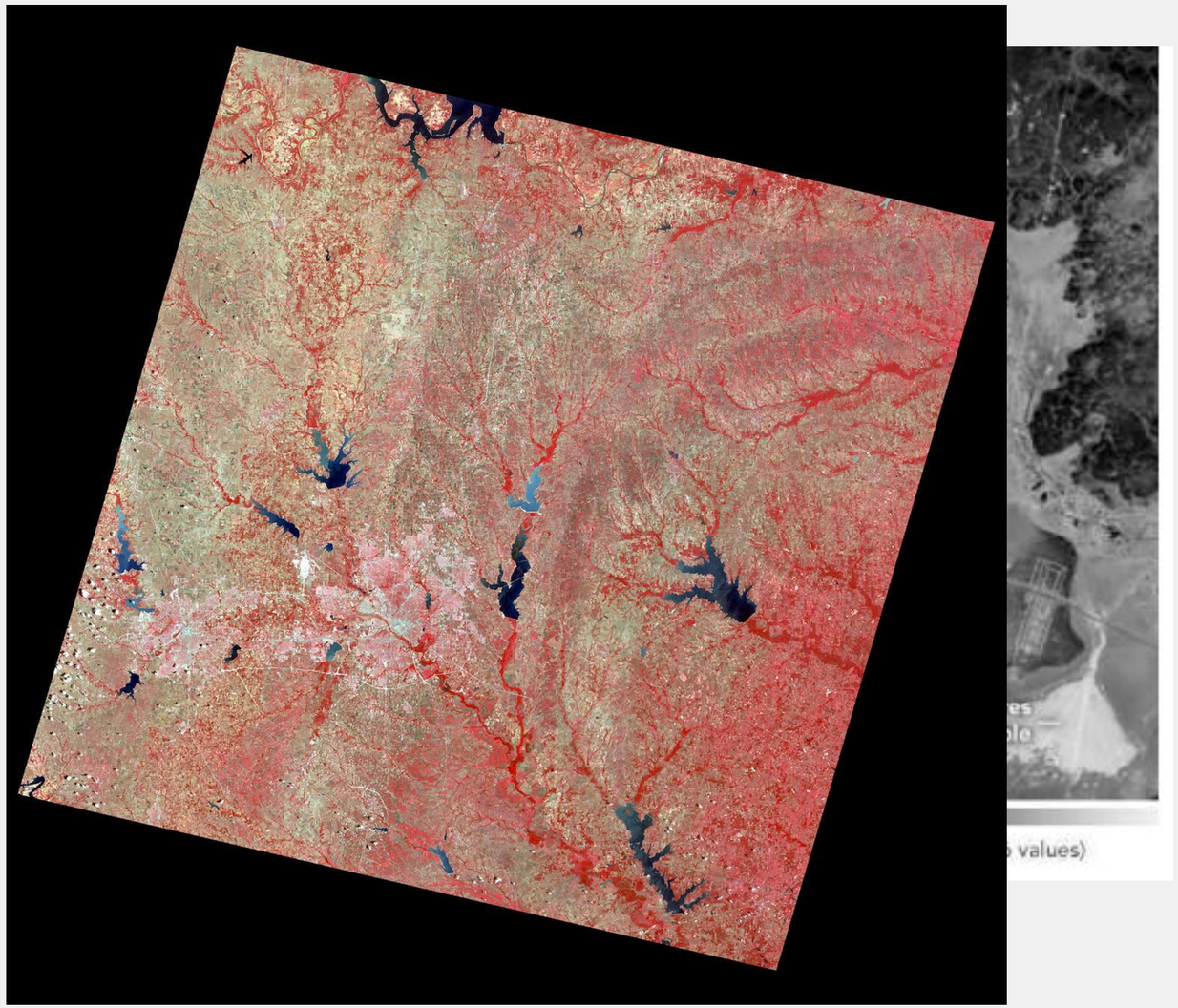
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Spectral resolution

- Describes a sensor's ability to discriminate differences in energy (or radiance)

- The better the radiometric resolution, the more sensitive the sensors is to small differences in energy. The larger number, the higher the radiometric resolution and sharper the image.

- 12-bit sensor, 4096 levels, Landsat & Sentinel-2
- 10-bit, 2024 levels: AVHRR
- 8-bit, 256 levels: Landsat TM
- 6-bit, 64 levels: Landsat MSS



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Advantages & disadvantages



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Advantages

Advantages

1. Large area coverage and cost effectively
- Can capture data across vast geographic regions
- Useful for monitoring country or continent-wide changes

2. Frequent revisit times
- Satellites enable time-series analysis and monitor changes

3. Access to remote or inaccessible areas
- Enables data collection in remote, dangerous or politically unstable areas

4. Data is freely available, and open access is growing rapidly
- Many countries and even commercial companies are opening their databases.

5. Satellite data supports many use cases and applications
- Agriculture, forestry, air quality, water quality, disaster management

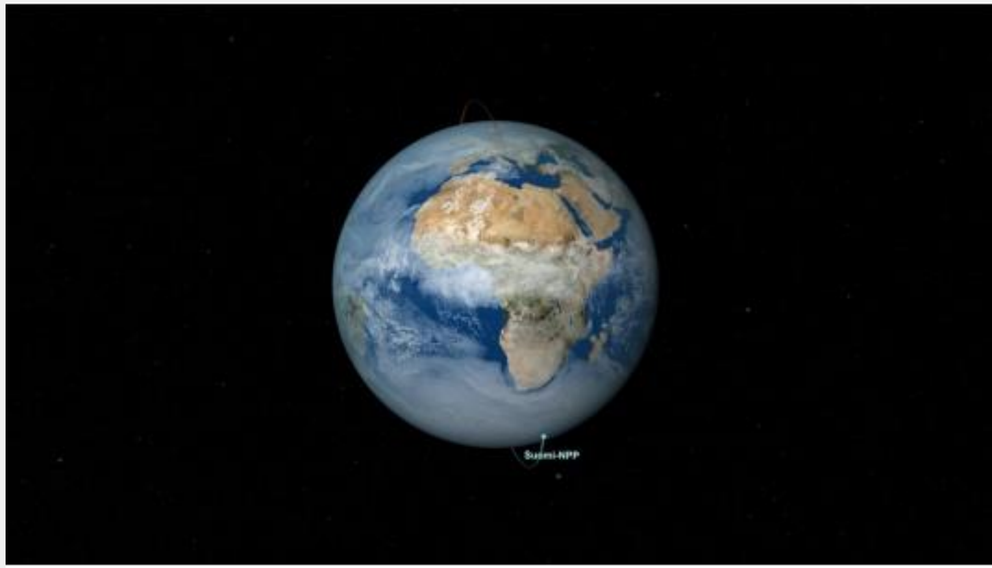


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Disadvantages

Disadvantages

1. Cloud obstruction: Earth is covered 80% in clouds high-resolution.
2. Very difficult to obtain high spectral, spatial, temporal and radiometric resolution at the same time
3. High cost for very high-resolution data ($ 1K:optical – $6K: SAR)
4. Analysis often requires ground truth which is time consuming and expensive to collect
5. Sensor and atmospheric effects obstruct the signal
6. Satellite data is big data! Requires heavy computational resources



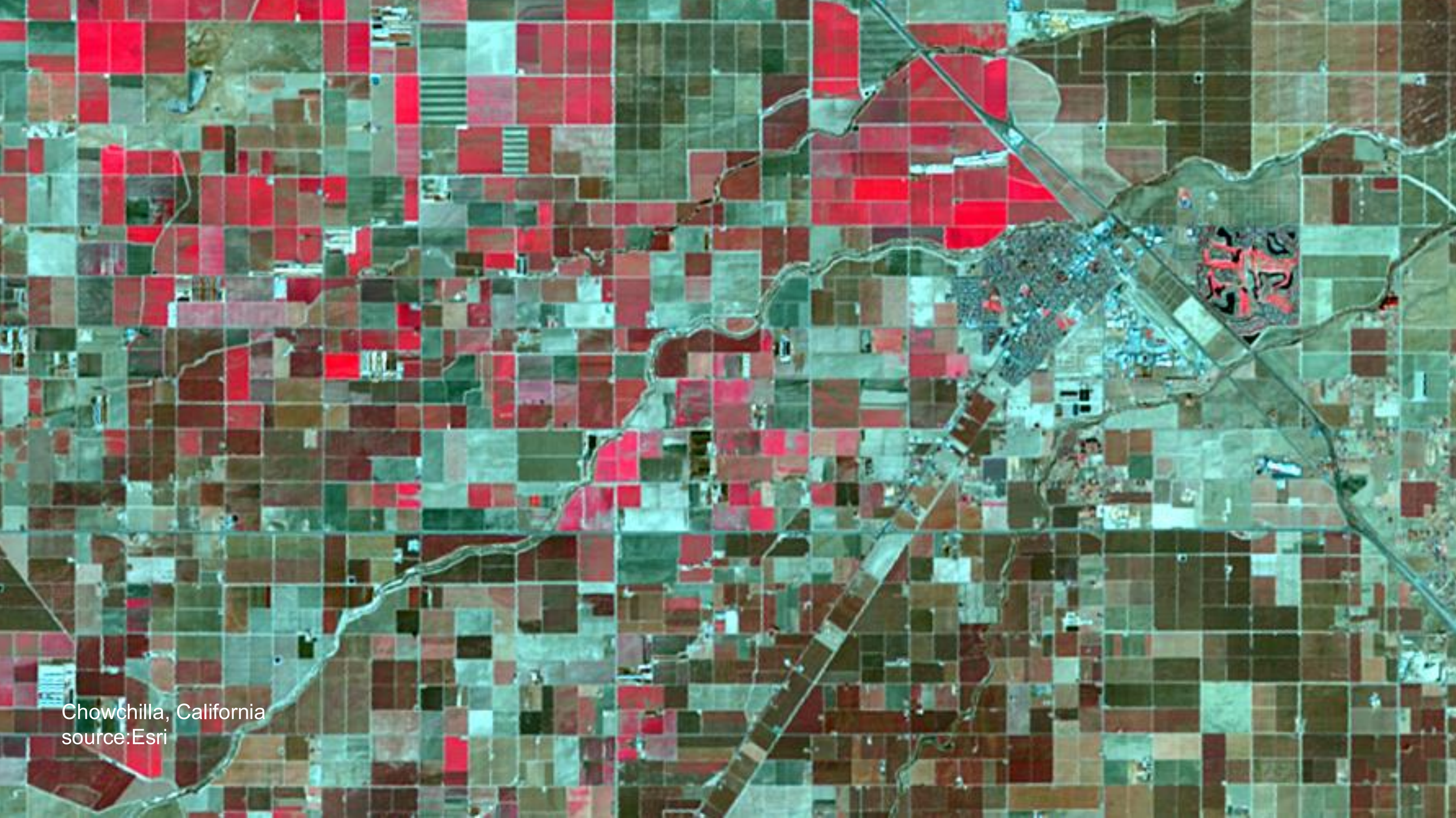
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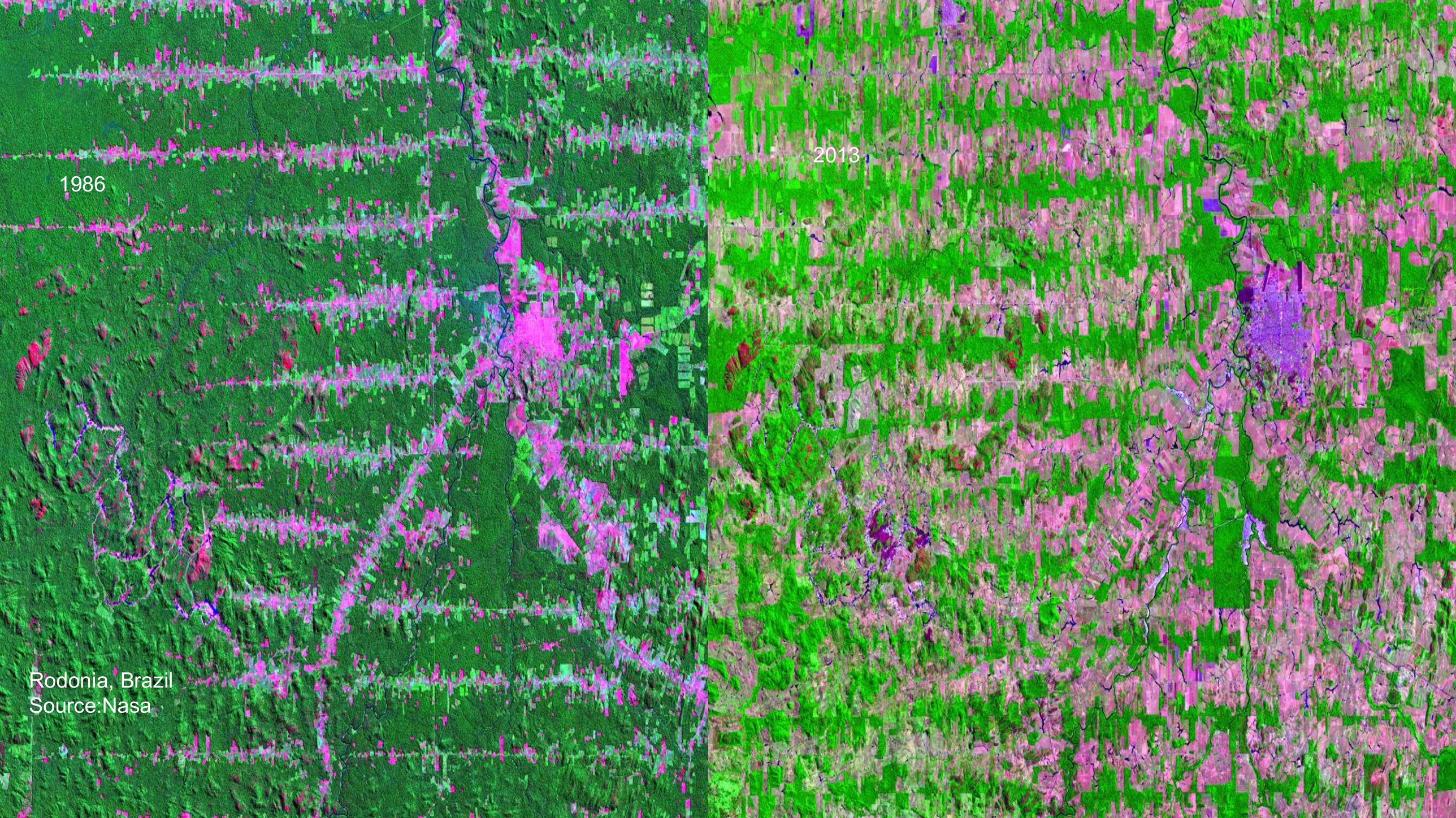
Applications



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Chowchilla, California
source:Esri

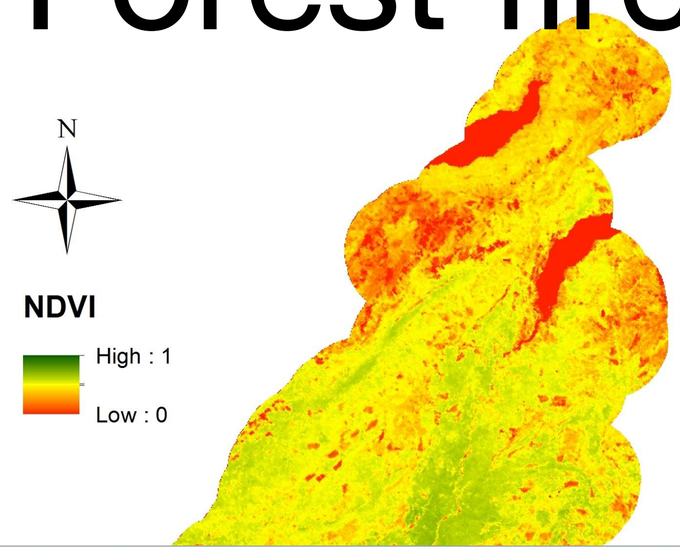

1986
2013
Rodonia, Brazil
Source:Nasa

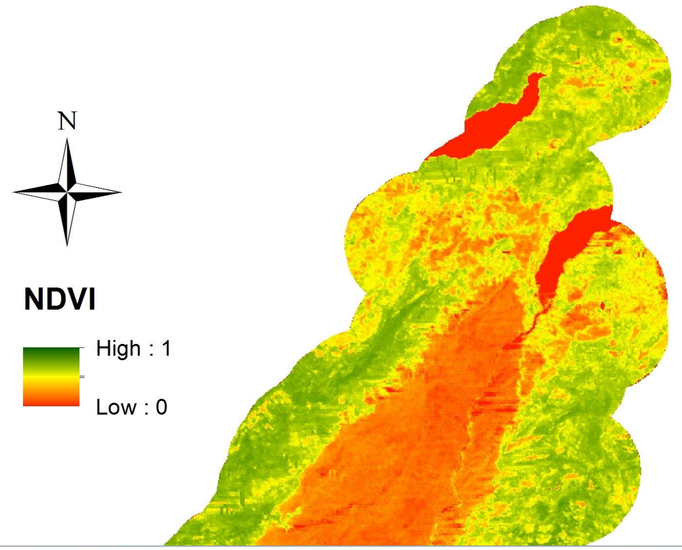
- Automatic Palm oil tree detection

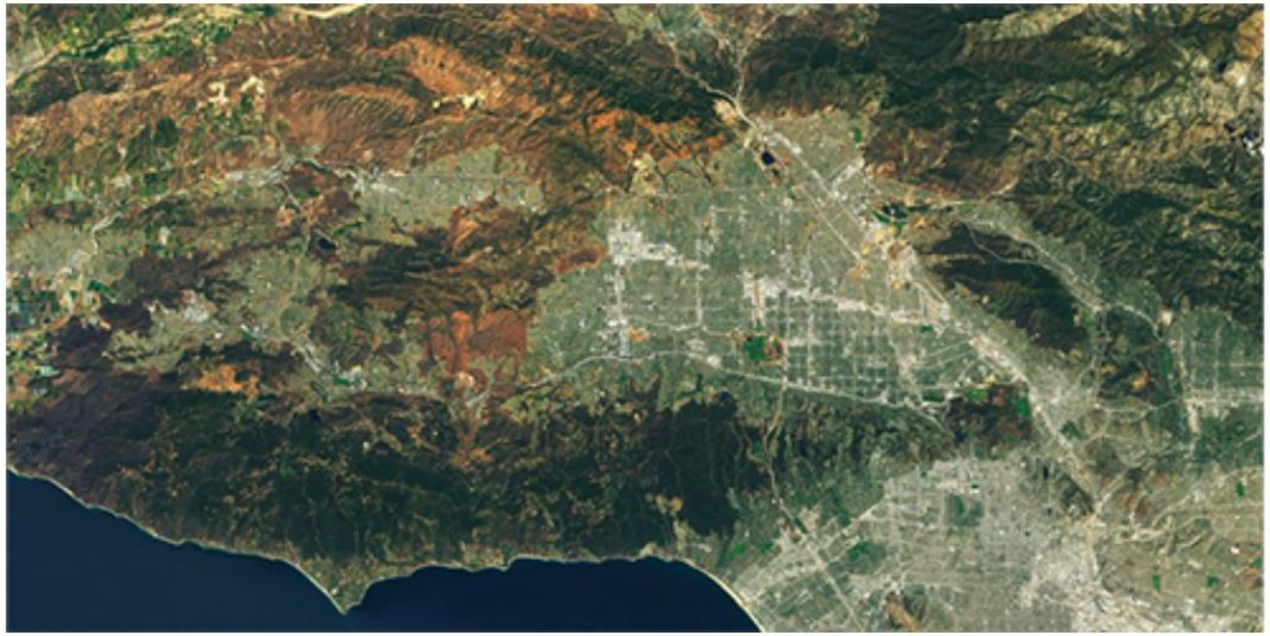


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Forest fires



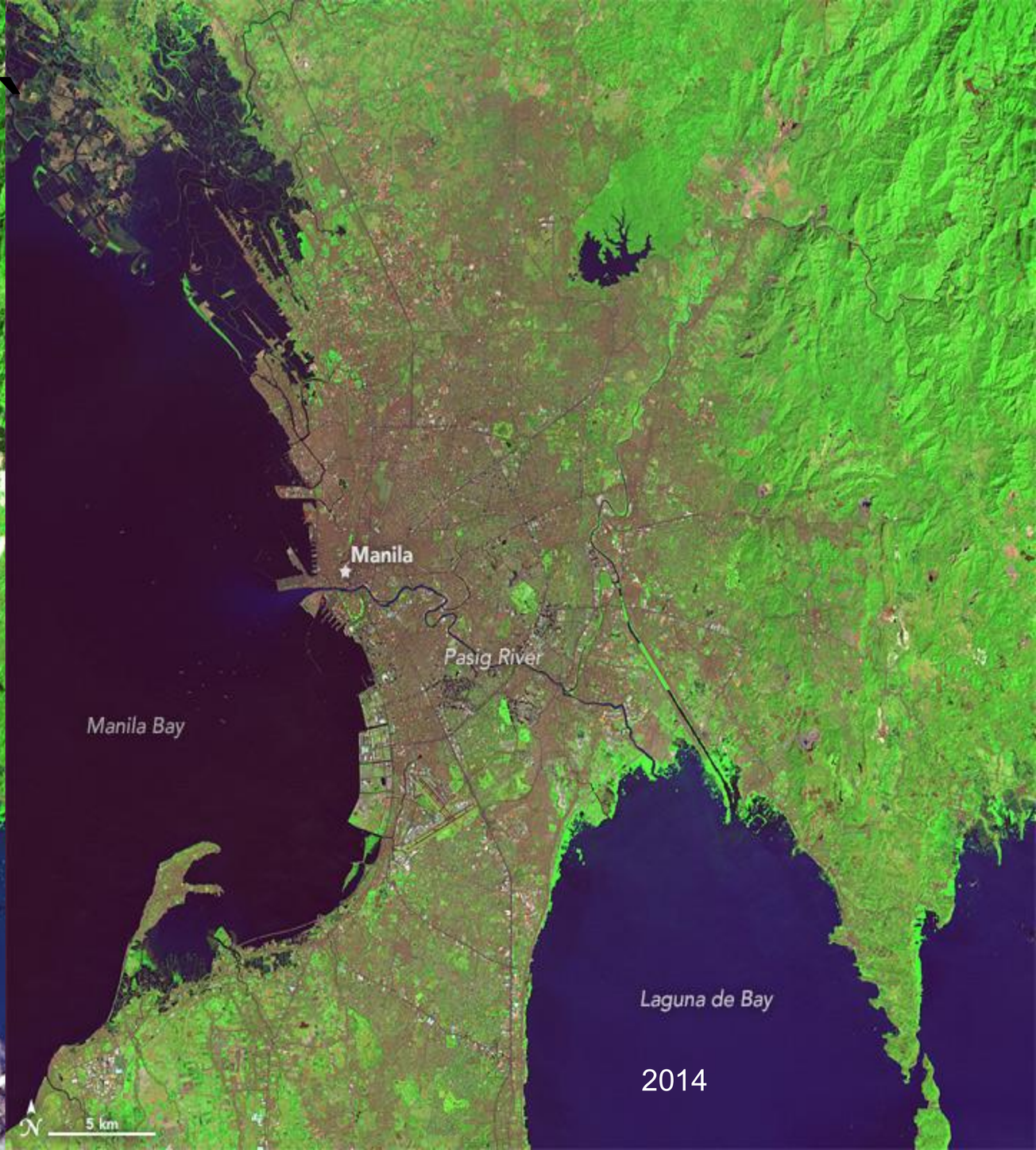


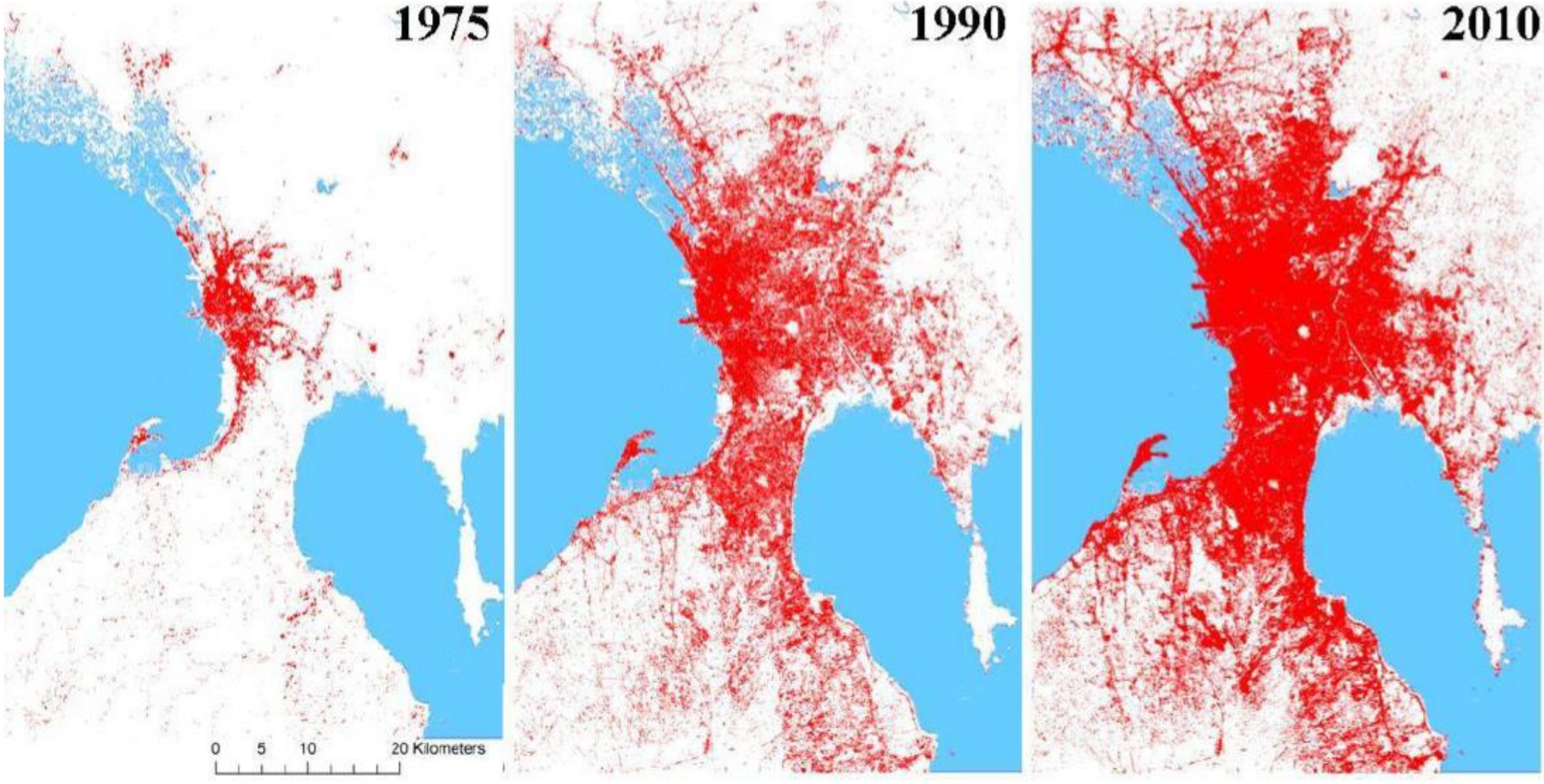




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Manila
Pasig River
Manila Bay
Laguna de Bay
1988
5 km


Manila
Pasig River
Manila Bay
Laguna de Bay
2014
5 km


1975
1990
2010
0 5 10 20 Kilometers

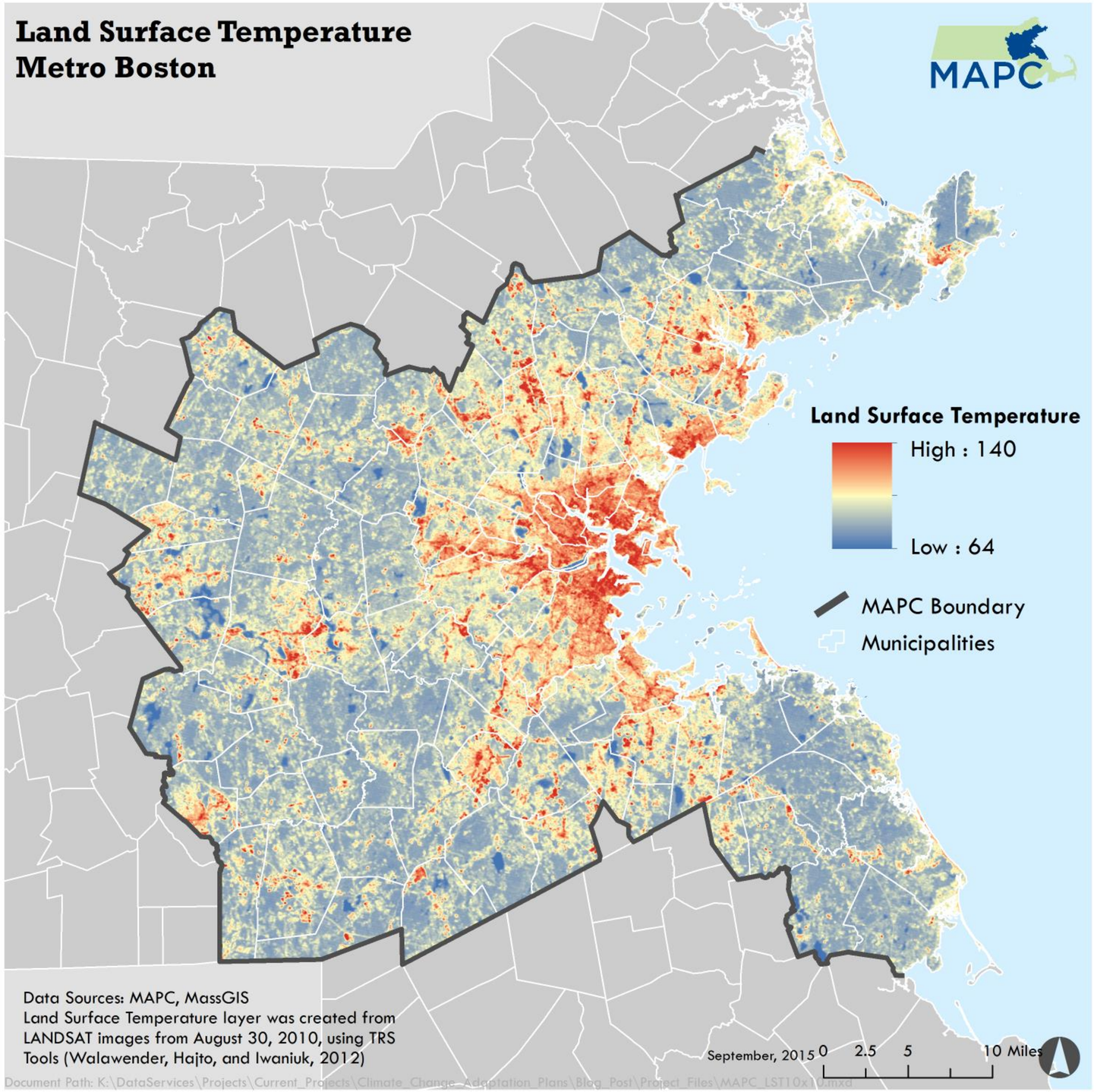




- classes
  - non-vegetation
    - bare land
    - building
    - roads
    - shadow
  - vegetation
    - grass
    - trees

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- Urban surface temperature
- Finding urban heat islands.
- Urban hot spots

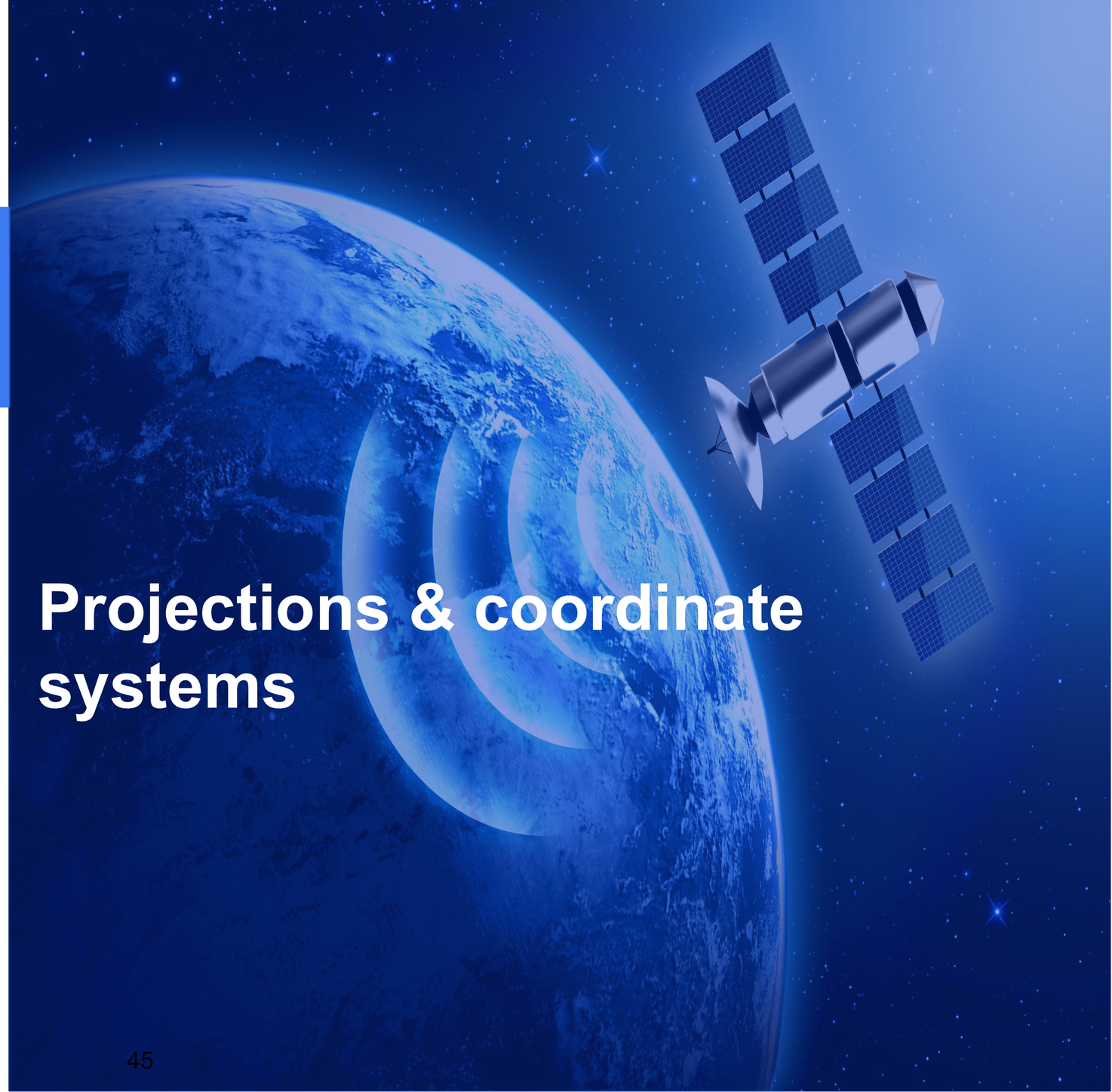


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Courtesy: DigitalGlobe
GEOSPATIAL WORLD
convertible
sedan
pickup truck
people

07

Projections & coordinate systems



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Earth's shape

The shape of the earth

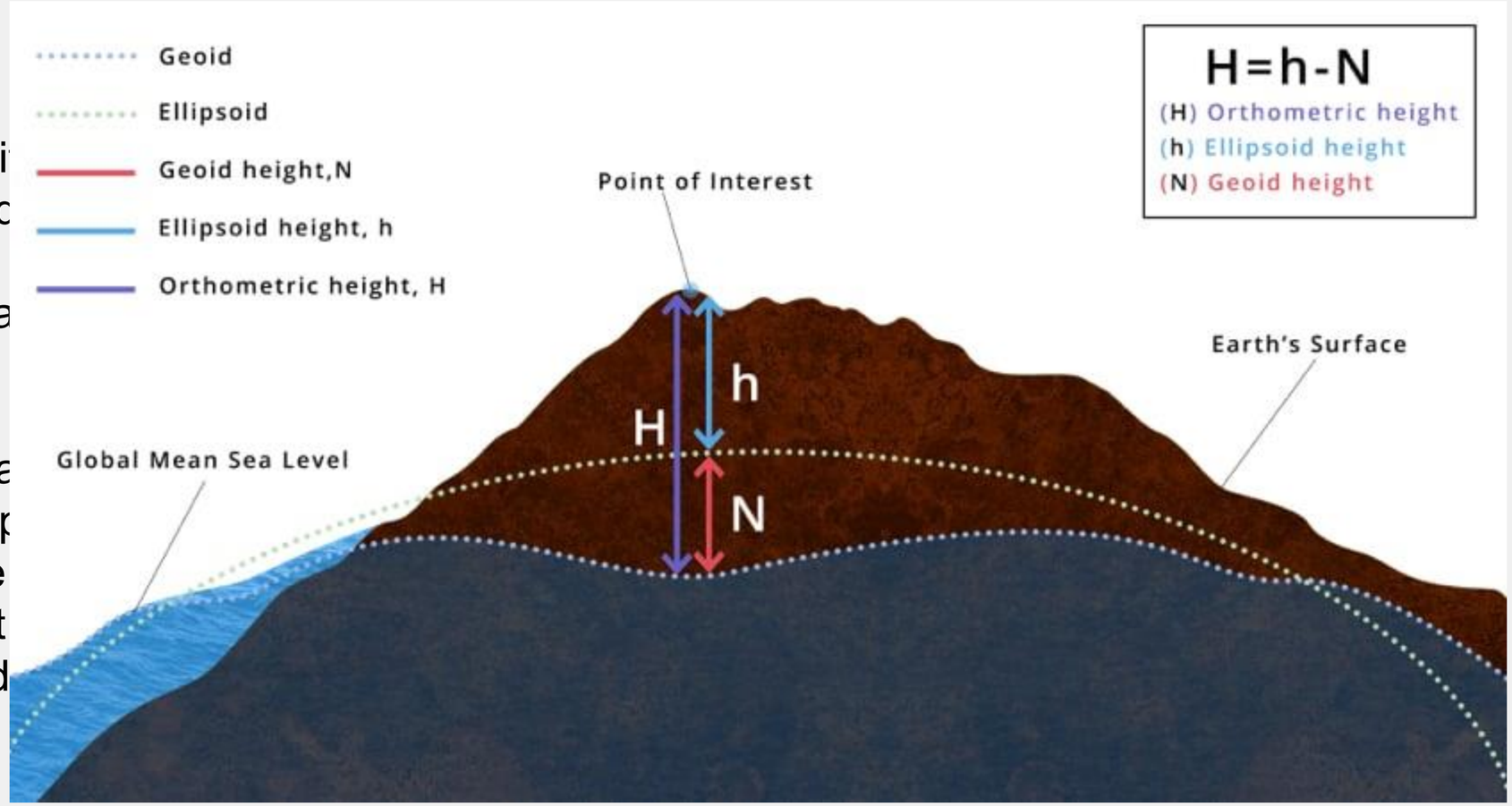
Earth is not what you think:

- It's not a perfect sphere, i
- The official term is: "geoic

- Geoid: A hypothetical sha
  mean level

To use spatial data we use a

- This gives us a global app
- For local analysis we use
- We use a reference point
- The Sir Bani Yas Island d



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Coordinate reference systems

All spatial data is refenced to a fixed point on earth

- Matching a pixel to a point's location on earth requires a **coordinate reference system (CRS)**.
- 2 types: geographic & projected

Geographic CRS (**WGS-84**)
- Good for global measurements or areas with large extents
- Can be less accurate for specific regions
- Bad in very precise areas

Projected reference systems (e.g. **UTM)**
- Works well for specific locations
- Good for areas with small spatial extent
- Accuracy can be skewed to one location, hard to compare.
- Does not work well with large areas

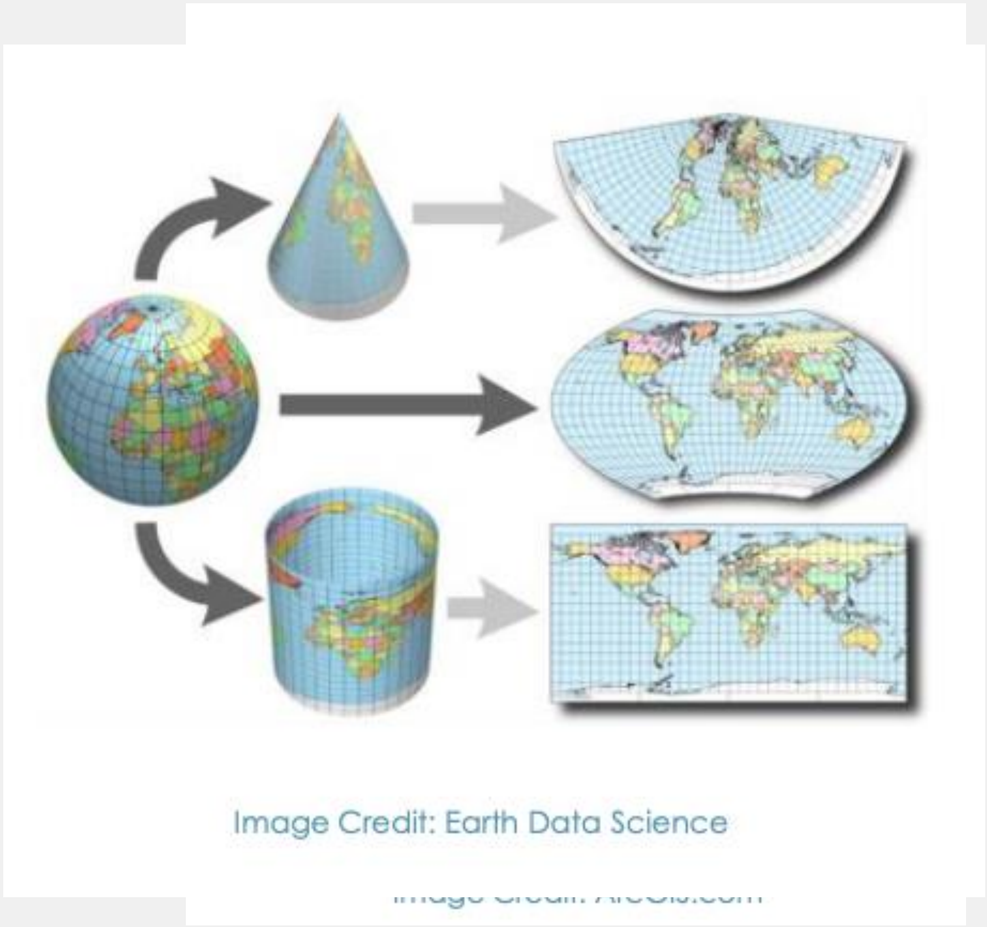


Image Credit: Earth Data Science

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08

Summary



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Summary

Remote sensing (**RS**): obtaining information about an object from a distance

Electro magnetic radiation is energy that travels in different wavelengths

Sensors can be active (e.g. radar, LiDAR) or passive (optical)

Spectral signature = an object's fingerprint

The application determines the data's temporal and spatial resolution requirements (global weather monitoring vs agriculture)



RS is a tool that can help humanity monitor the earth's changes and is a low-cost tool

Earth =/= a sphere but a potato 🥔

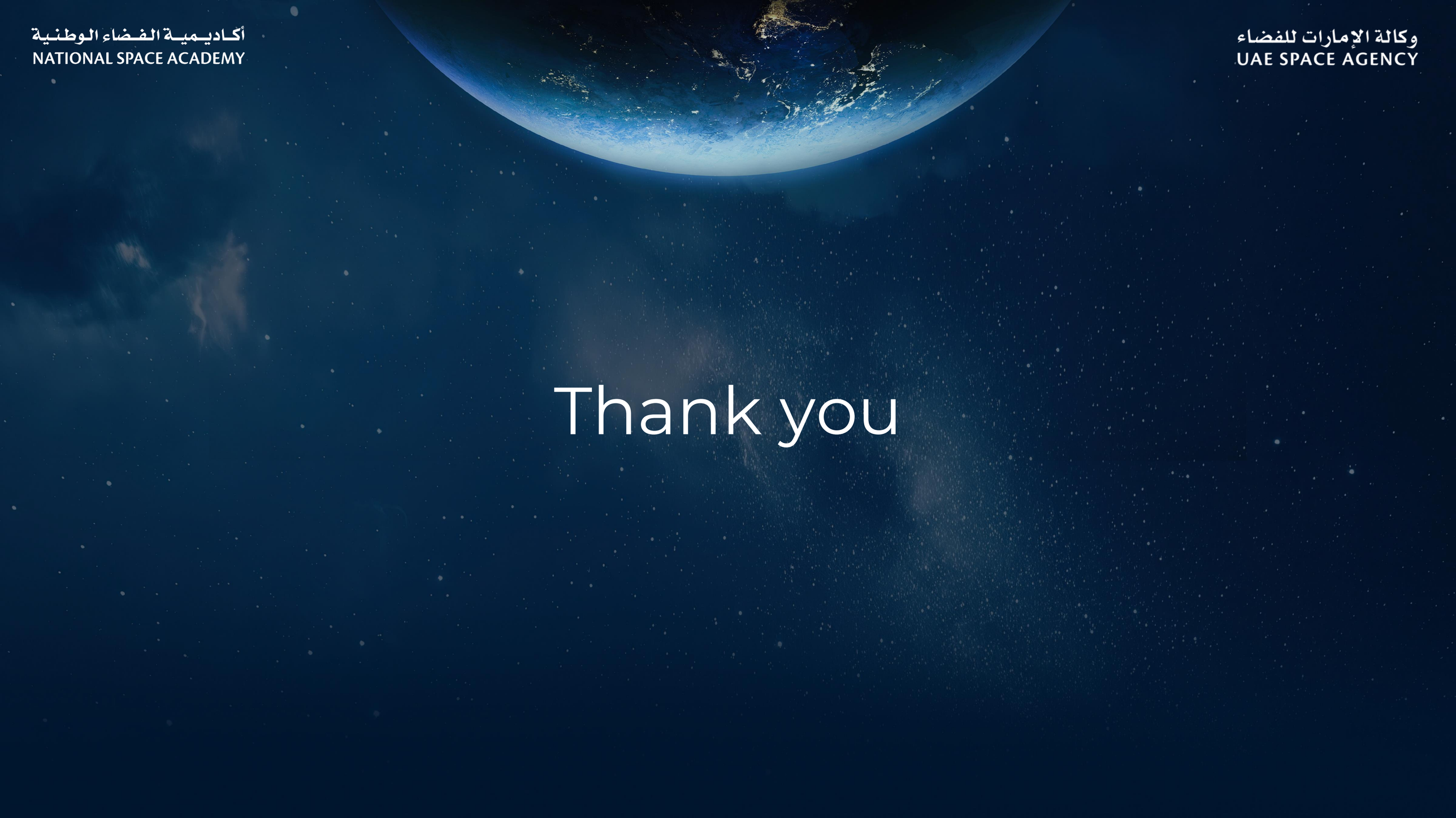
Two types of coordinate reference systems:

- Geographic CRS (**WGS-84**) = for large areas
- Projected reference systems (e.g. **UTM)** = works well in precise, small areas

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UAE SPACE AGENCY



Thank you